Heads in the Sand: Theory and Experiment on Information Avoidance in Groups

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PRELIMINARY AND INCOMPLETE

Abstract

A growing literature shows that individuals often avoid freely available information, even when it is instrumentally valuable. While well-documented in individual settings, less is known about how such choices operate in groups, where they generate negative externalities. Such externalities endogeneize the prospects that agents face, potentially affecting others' willingness to learn potentially bad news. We investigate whether information avoidance is interdependent: does the expectation that others will remain ignorant increase one's likelihood of avoidance? Our theoretical model predicts individuals can be classified into distinct strategy types: some exhibit strategic complementarity ("contagious ignorance"), others strategic substitutability, while some are unconditional avoiders or acquirers. We test the prediction of interdependence with a novel laboratory experiment where participants choose whether to learn a binary state of the world, which determines exposure to an aversive event and whose negative impact is mitigated by the group's aggregate information acquisition. To assess interdependence, we exogenously vary beliefs about others' avoidance and use the strategy method to elicit full reaction functions. We find that, on average, information avoidance is contagious: leading subjects to believe that the majority (vs. the minority) of peers avoid information increases the avoidance rate by a third. This effect appears to be mediated by a heightened reported aversion to discovering bad news, with no corresponding role for utility gains from discovering good news. We uncover significant heterogeneity behind the average contagiousness: only a subset of individuals ("Strategic Complements") drive this contagion, while "Strategic Substitutes" respond in the opposite direction. Simulations show that this heterogeneity leads to nonlinearities in how group composition determines equilibrium outcomes, including the risk of collective ignorance. We also find that prosocial preferences moderate these decisions, with altruism reducing avoidance. Our results show that willful ignorance can be contagious among people, but the aggregate equilibrium take up of information critically depends on the distribution of individual strategy types within a group.

1 Introduction

A fundamental assumption in Economics is that information is a valuable resource for decision-making, leading to the prediction that rational agents will typically acquire costless, instrumental information to improve their outcomes (Stigler, 1961). However, a growing theoretical and empirical literature challenges this view, documenting that individuals often exhibit intrinsic preferences for information —deriving psychological utility from information and resulting beliefs beyond its instrumental value— and sometimes choose to avoid freely available knowledge about payoff-relevant states (e.g., their own health status or the quality of their financial assets), even when this hinders subsequent decision-making (Oster et al., 2013; Ho et al., 2021; Falk and Zimmermann, 2024).[1]

While this phenomenon of "information avoidance" is well-documented in individual decision-making, much less is known about how such decisions operate in group settings, where individual decisions to remain ignorant about the state of the world may generate *negative payoff externalities* on others. In many collective scenarios (from mitigating climate change and containing infectious diseases to ensuring a company's financial health) an individual's decision to remain ignorant about the state prevents her from taking optimal mitigating actions (e.g., adopting eco-friendly measures, using masks and vaccinating, and divesting from toxic assets), thereby imposing a negative externality on others in the group (e.g., larger climate catastrophe, increased probability of infections, higher likelihood of bankrupcy and unemployment). Consequently, the *stakes* or *prospects* faced by each individual are *endogenous* to the information decisions of other members. Crucially, if an individual expects other members to remain ignorant about the state, her own prospects in the bad state worsen due to the inaction of others. This endogenous downfall of prospects

[1] For a comprehensive review, see Golman et al. (2017)

can make the possible event of discovering bad news about the state more psychologically terrifying, increasing her own incentive to avoid information. This dynamic can lead to a self-reinforcing equilibrium of collective ignorance, where individuals, like "ostriches" burying heads in the sand, fail to acknowledge the crisis and to coordinate on its mitigation. By remaining ignorant and inactive, members make revelation of a bad state more dire, thereby promoting generalized ignorance, inaction, and failure to address looming collective crises. As a result, sometimes individuals collectively resist acquiring necessary information, despite it being readily available. A key question, therefore, is whether information avoidance is *interdependent*: does the expectation that others will avoid information (and therefore, make prospects worse) make an individual more likely to do so herself? In other words, can willful ignorance be contagious? A second, related question concerns the role of social preferences: since information decisions generate externalities on other members, do prosocial motives (such as altruism and reciprocity) mitigate or amplify this interdependence?

This paper answers these questions by combining theory, experiment, and simulations. Our investigation begins with a theoretical model inspired by Bénabou (2013), which studies the strategic interdependencies that may arise in group settings. As well as contagious ignorance, we show that in our setup there is also an alternative possibility. Depending on the nature of their utility function, individuals may exhibit strategic substitutability in information avoidance: when others bury their heads in the sand, these individuals become less inclined to shun information. These findings indicate that the dynamics of information avoidance in groups may vary considerably. Furthermore, the theory also shows that some individuals may not base their choices on the actions of others at all, and may instead always avoid or always acquire information, regardless of what others do.

Our model also considers the role of social preferences. Since information acquisition involves a positive payoff externality, more altruistic individuals (who care more about the welfare of others) might be expected to be more likely to become informed. However, our model shows this is not necessarily the case. A more altruistic agent internalizes the welfare of others more strongly, which can make discovering a bad state feel even worse. Therefore, an overproportionate aversion to bad news relative to good news (or, equivalently, a dislike for variance in the lotteries over beliefs induced by learning the state) can lead an altruistic agent to avoid information where a self-interested agent would acquire it. In addition to altruism, we also consider how reciprocity concerns motivate choices. Since an agent benefits from the positive payoff externalities generated when others acquire information and is harmed by the negative externalities when others avoid it, positive and negative reciprocity may induce the agent to match the actions of others.

To closely examine how information avoidance decisions are influenced by the choices of others in practice, we design and conduct a novel lab experiment. In the experiment, participants are randomly divided in groups and are informed that there are two possible states of the world: *Good* and *Bad*. In the *Bad* state, everyone in the group will be exposed to aversive stimuli: they will hear distressing screams through their headphones in the second part of the experiment. In the *Good* state, no noise will be heard. At the beginning of the experiment, we ask each participant to choose whether they want to discover the state immediately ("Now" option) or if they prefer to wait until the state is revealed ("Later" option). If the state is *Bad*, the volume of the screams will be reduced if more individuals choose Now. In other words, each participant understands that by selecting Now, they can lessen the intensity of the noise for both themselves and everyone else in their group, should the state turn out to be *Bad*.

Our approach to examining how individuals respond to the decisions of others is twofold. First, we exogenously vary the subjects' expectations about the share of information avoiders among their peers by providing information about the corresponding share observed in a pilot experiment. We implement two treatments, randomized at the level of experimental groups: the *Many Avoided* treatment, in which subjects are informed that 80% of pilot participants avoided information; and the *Few Avoided* treatment, in which subjects are informed that the share is much smaller, only 20%. By comparing the rate of avoidance among participants in the *Many Avoided* and *Few Avoided* treatments, we can measure how a shift in expectations about the share of avoiders among others (and hence, about the severity of the outcome in the Bad state) influences decisions to acquire or avoid information, on average, at the group level. Second, we use the strategy method (Selten, 1967; Fischbacher et al., 2001) to examine how each participant reacts to varying levels of information avoidance within their group. This allows us to uncover potential heterogeneity in individual reaction patterns.

The data indicate that our treatment successfully influences the participants' beliefs about the likely incidence of information avoidance among their peers, in the expected direction. In turn, this affects the participants' choices. In the *Many Avoided* treatment, the share of participants who avoid information increases by approximately a third relative to the *Few Avoided* treatment (from 24% to 33%). This establishes that, at the aggregate level, the data are consistent with contagious ignorance. We show that, on average, the *Many Avoided* condition worsens the anticipatory utility participants believe they will experience if they find out that the state is Bad. We also find evidence that, in our group setting, social preferences matter for information avoidance. Ceteris paribus, more altruistic individuals are less likely to avoid information.

To shed light on mechanisms and to better understand determinants of the choice of avoiding information, we develop a novel elicitation of anticipatory utilities. In particular, we elicit the anticipatory utilities that subjects expect they will incur *conditional on each information set* —specifically, upon acquiring information and finding out good news about the state, upon acquiring information and finding out bad news, and upon acquiring no news at all. The elicitation, which maps the theoretical components of the anticipatory utility as specified in the model (developed in Section 3), allows us to empirically decompose the net (anticipatory) incentives to avoid information, and to study through which components of such incentives the treatment operates. We find that in the *Many Avoided* condition —where subjects are exogenously led to believe that many others in their group will avoid information, and therefore that screams in the bad state will be more severe— subjects expect more anticipatory *dis*utility of finding out bad news relative to subjects in the *Few Avoided* condition, but there are no differences across conditions in the expectation of anticipatory utility upon finding good news nor upon getting no news. This pattern lends support to the groupthink mechanism, as distinct from other, more traditional mechanisms (such as conformity and herding).

Turning now to the individual-level data, we uncover significant heterogeneity in how participants respond to varying incidences of information avoidance in their group. Among those who are influenced by the choices of others (about 40% of the total), we find that half exhibit a pattern consistent with contagious ignorance (strategic complementarity), while the other half shows the opposite tendency (strategic substitutability). There is also a substantial share of participants whose information avoidance choice is independent of their teammates' decisions. The larger portion consists of those who choose to always acquire information (approximately 42%), while just under 9% of participants consistently choose to avoid information.

The heterogeneity in individual reaction functions underscores the importance of group composition for the aggregate incidence of information avoidance. To trace the relationship between group composition and the aggregate incidence of information avoidance, we employ simulations based on the empirical distribution of reaction functions. Groups primarily composed of individuals who exhibit strategic complementarity in information acquisition are at risk of remaining trapped in a so-called Mutually Assured Ignorance (MAI) equilibrium (see Section 3), where no one acquires information. The presence of unconditional information avoiders can precipitate this outcome, pushing the group toward the MAI equilibrium. Similarly, the presence of unconditional information getters may drive the group toward a virtuous equilibrium where everybody acquires information (Mutually Assured Awareness equilibrium). In contrast, groups composed of individuals who exhibit strategic substitutability never reach a MAI equilibrium, but they are also unable to sustain an equilibrium characterized by universal information acquisition. Our simulations suggest that there are important nonlinearities in the relationship between group composition and the equilibrium incidence of information avoidance: the same, fixed amount of change in group composition (e.g., a 10 percentage point increase of unconditional information avoiders) leads to different amounts of increase in the equilibrium level of avoidance.

Our findings have implications for the design of organizations, in particular for the screening of personnel and organizational processes. In particular, organizations may want to assemble teams in such a way that limits the probability of a MAI equilibrium. Second, prosocial preferences are another important dimension, as these can naturally mitigate the avoidance of information, minimizing the scope of a groupthink equilibrium. A similar argument applies to general preferences for information, which could be measured, for example, with the Ho et al. (2021)'s scale (which, we find, is correlated with information avoidance in our experiment).

The remainder of the paper is organized as follows. In Section 2, we discuss related literature. In Section 3, we lay out our theoretical framework. In Section 4, we present the experimental design. In Section 5, we discuss the empirical findings. In Section 6, we capitalize on our data to calibrate how different group compositions may affect equilibrium outcomes. Finally, in Section 7, we conclude.

2 Related Literature

Our study relates mainly to two strands of literature: attitudes towards information and suboptimal decision-making in groups.

First, we contribute to the literature on attitudes towards information from both theoretical and empirical perspectives. On the theoretical side, standard economic theory recognizes that information is a valuable resource for decision-making and accordingly predicts that, if available at no cost, agents will acquire information about payoff-relevant states of the world to improve their subsequent decisions. However, a growing literature documents and acknowledges the possibility that economic agents may have intrinsic preferences for information beyond and above the instrumental, decision-making value of information. Starting from Kreps and Porteus (1978), the theoretical literature has generated a variety of models in which agents may prefer to remain uninformed, or delay the acquisition of information, despite information being available at no or low cost. Golman et al. (2017) provides a review of models of information avoidance, defined as the deliberate choice to avoid freely available information when the agent is aware of its existence. Bénabou (2013), the model closest to our work, extends the analysis of information avoidance decisions to group settings. It shows theoretically that when willful blindness to information makes other agents worse off, it can become contagious. This rises the possibility of multiple equilibria, including a pervasive one in which all agents remain ignorant. We contribute to this literature by empirically examining whether, as predicted by theory, information avoidance exhibits interdependences in groups. In addition, we theoretically examine how prosocial preferences affect incentives to avoid information in groups where remaining ignorant makes other agents worse off.

On the empirical side, several studies both in the lab and in the field documented that individuals often prefer to remain ignorant despite information being available at no or low cost and being useful for subsequent decision-making (e.g., Dana et al., 2007; Karlsson et al., 2009; Oster et al., 2013; Ganguly and Tasoff, 2017; Pagel, 2018; Ho et al., 2021; Meissner and Pfeiffer, 2022; Masatlioĝlu et al., 2023; Momsen and Ohndorf, 2023; Engelmann et al., 2024; Falk and Zimmermann, 2024). Across studies and contexts, rates of avoidance range from 5% (Ganguly & Tasoff, 2017) to more than 90% (Oster et al., 2013). The vast majority of empirical studies have focused on information avoidance in individual decision-making settings. We contribute by documenting that, in groups, a substantial proportion of individuals engage in information avoidance, even when it worsens the future outcomes for both themselves and other members of their group.

Our paper is most closely related to Falk and Zimmermann's (2024) work, which studies intrinsic preferences for information about a potentially adverse future event, in an individual setting. In their experiment, information acquisition generates no material returns: the decision-maker's future payoff is independent of whether information is acquired. We build on their experimental design to study information avoidance decisions in groups. In addition, we introduce a positive return of information acquisition. We show that, when remaining uninformed makes other people worse off, information avoidance decisions can be interdependent. In addition, we examine how prosocial preferences affect the decision to acquire information.

Second, our study also relates to the literature on non-Bayesian information processing, overconfidence, and suboptimal decision-making in groups. A recent literature suggests that decisions in groups can become correlated through the social transmission of biases. Suvorov et al. (2024) shows that members of a group share noisy feedback about their own ability to other group members in a selective and asymmetric way: group members are more likely to share positive feedback about their ability to perform in an investment task than negative feedback. This selective information sharing leads to groups becoming overconfident about the group's ability to perform in investment decisions, and this overconfidence leads to suboptimal investment decisions by low ability groups. A closely related study by Oprea and Yuksel (2022) shows that when individuals can socially exchange ego-relevant beliefs, their initial biases amplify because subjects respond to their counterparts' beliefs in an asymmetric way. We show that groups can also attain suboptimal outcomes by deliberately avoiding learning from freely available signals, when signals reveal the state perfectly and are not ego-relevant, and without communication among group members.

3 Theoretical Framework

We now describe our theoretical framework and key results. All proofs are in the Appendix. Consider the following setup, inspired by Bénabou (2013). There are three periods, date 0, 1 and 2 and two states of the world, B (bad), occurring with probability $p \in (0, 1)$ and G (good), occurring with probability $1 - p$. Material payoffs are received at date 2 and generate utility u_i. At date 1, agents evaluate lotteries over date-2 outcomes according to the expected utility function $U_i = E[u_i]$. At date 0, agents evaluate lotteries over date-1 utilities U_i according to the expected utility function $E[v_i(U)]$, where $v_i(.)$ is a strictly increasing function capturing i's preferences over expected utility lotteries in the spirit of Kreps-Porteus (1978). Denote with g the utility obtained at date 2 when the state of the world is good and with b when the state of the world is bad. b depends on whether an individual chooses to acquire or to avoid information about the state of the world at date 0. Intuitively, individuals who are aware that the state is bad can take corrective measures to reduce its negative consequences, whereas those who remain unaware cannot.[2] Thus, the utility at date 2 when the state is bad is equal to

$$b \equiv \bar{b} - \lambda_{-i} - \Delta \text{ if } i \text{ avoids information}$$

and

$$b + \Delta = \bar{b} - \lambda_{-i} > b \text{ if } i \text{ acquires information}$$

where $\bar{b} < g$ is the payoff when the state of the world is bad and all group members acquire information (and undertake corrective measures as a result), $\lambda_{-i} \in [0, \frac{n-1}{n}]$ is the share of group members other than i who avoid information, n is the number of individuals in the group and $\Delta \equiv \frac{1}{n}$. In words, Δ captures the instrumental value of acquiring information. It reflects how much an agent can lessen the adverse effects of a bad state by obtaining information and taking appropriate corrective measures. Note that, when the state of the world is bad, utility is decreasing in λ_{-i}, the share of i's fellow group members who avoid information. By avoiding information, agents not only damage themselves but also exert a negative externality on others.

In what follows, we assume that

$$(1-p)(g-\bar{b}) > \Delta. \tag{1}$$

This assumption is not necessary for our results but simplifies the exposition. In words, it ensures that the expected value of a lottery where i obtains $\bar{b}$ with probability p and g with probability $1-p$ exceeds the value of obtaining $\bar{b} + \Delta$ for sure.[3]

An individual's date 0 *net* expected utility from avoiding information is

$$\phi_i \equiv v_i(pb + (1-p)g) - [pv_i(b+\Delta) + (1-p)v_i(g)] \tag{2}$$

where $v_i(pb + (1-p)g)$ is the date-0 expected anticipatory utility from avoiding and $pv_i(b+\Delta) + (1-p)v_i(g)$ is the expected anticipatory utility from acquiring information. If expression (2) is positive, individual i strictly prefers to avoid information at date 0, while if expression (2) is negative, individual i strictly prefers to acquire it.

[2]This is imposed exogenously for simplicity but could arise endogenously in a setup where taking corrective action is costly, so that under ignorance agents would choose not to exert any corrective measure.

[3]Formally, $p\bar{b} + (1-p)g > \bar{b} + \Delta$.

3.1 Types of Best Response ("Strategy Types")

We now explore how the nature of v_i determines an individual's optimal information acquisition choice as a function of others' choices (i.e., her best reaction function). The first two results consider v_i that is everywhere convex or everywhere concave.

Lemma 1 *If v_i is everywhere convex, individual i is an "Always Getter". At date 0, she acquires information independently of the information acquisition choices of other agents in her group.*

If v_i is convex, the utility boost obtained from discovering that the state is good outweighs the utility drop when the state is bad. Consequently, the agent always prefers to acquire information. Consider now the case where v_i is concave, as in Bénabou (2013).

Lemma 2 (Bénabou, 2013) *If v_i is everywhere concave, individual i (i) is an Always Getter (ii) is an "Always Avoider" or (iii) is affected by the choices of others in her group in the following way: there exists an interior threshold value λ_i^* such that i strictly prefers to avoid information if $\lambda_{-i} > \lambda_i^*$ and strictly prefers to acquire information if $\lambda_{-i} < \lambda_i^*$ (Strategic Complementarity).*

In the Appendix, we characterize the necessary and sufficient conditions for each of (i), (ii) or (iii) to apply. Always Getters and Always Avoiders make the same information acquisition choice independently of the choices of others in their group. Alternatively, the agent may condition her decision on the choices of others in her group, exhibiting strategic complementarity. The intuition for the result is as follows. Without the instrumental value of information, the agent would always prefer ignorance, since the utility loss from discovering the bad state outweighs the gain from finding out that the state is good (this follows directly from concavity). In the presence of instrumental value, as in our setup, the agent may opt to acquire information. However, as more agents choose to avoid information, the outcome in the bad state deteriorates, increasing the utility loss associated with discovering the bad state. As a result, meeting the conditions for information acquisition becomes increasingly challenging. This pattern may give rise to strategic complementarity in the agent's information acquisition strategy. The agent acquires information only if a sufficient share of her peers also do so; otherwise, she avoids information.

Consider now a reference dependent v_i (Kahneman & Tversky, 1979; Tversky & Kahneman, 1992). The agent's utility is largely insensitive to expected payoffs except around some reference level. Denoting i's reference expected utility as $\widehat{U_i}$, we have (i) $v_i''(U) < 0$ for $U > \widehat{U_i}$ and (ii) $v_i''(U) > 0$ for $U < \widehat{U_i}$.

Lemma 3 *If v_i is reference dependent, under some conditions the following applies: there exists an interior threshold value λ_i^{**} such that i strictly prefers to avoid information if $\lambda_{-i} < \lambda_i^{**}$ and strictly prefers to acquire information if $\lambda_{-i} > \lambda_i^{**}$ (Strategic Substitutability).*

In the Appendix, we provide sufficient conditions for strategic substitutability. To illustrate, assume that if the state is known to be bad utility is always below the reference level while it is above it in the good state. First, suppose that many agents in i's group avoid information. The final payoff in the bad state is so low that, at date 1, the expected utility under ignorance falls below the reference level. Relative to remaining ignorant, the utility loss from discovering the bad state is minimal (since both situations generate utility that is below the reference level), while the gain from discovering the good state is large. As a result, the agent prefers to acquire information. Now, suppose that many agents acquire information. The final payoff in the bad state is not too low and the expected utility under ignorance is above the reference level. In this case, under mild conditions, the agent prefers to avoid information, since the utility gain from discovering the good state is minimal relative to the utility loss from discovering the bad state.[4]

The following proposition summarizes our results.

Proposition 1 *Depending on the nature of v_i, i's propensity to avoid information may be (i) independent of, (ii) increasing (strategic complementarity) or (iii) decreasing (strategic substitutability) in the share of group members who avoid information.*

In what follows, we will use the term "strategy type" to indicate whether an individual is an Always Getter, an Always Avoider, or alternatively exhibits preferences featuring strategic complementarity (in short, is a "Complement") or substitutability (is a "Substitute").

Naturally, the specific information avoidance equilibria that may be reached vary depending on the agents' strategy types. Bénabou (2013), for example, examines the risks that emerge when agents are Complements. In that case, groups may become trapped in a "Mutually Assured Ignorance" (MAI) equilibrium where no one seeks out information, and as a result, no corrective actions are taken to shield the group from adverse outcomes in the face of a bad state of the world —potentially with disastrous consequences.[5] By the

[4]For example, suppose $b(\lambda_{-i} = 0) = -10$, $b(1) = -90$, $g = 50$, $\Delta = 5$, $p = 0.5$, $\widehat{U}_i = 0$, and $v_i(U) = U^{1/3}$ if $U > 0$ and $= -(-U)^{1/3}$ if $U < 0$. Then $\varphi^i(0) \approx 2.58$ and $\varphi^i(1) \approx -2.87$, so the agent exhibits strategic substitutability.

[5]Bénabou (2013) uses the acronym MAD (Mutually Assured Delusion) equilibrium. However, in the context of information avoidance MAI provides a more accurate description of the equilibrium. The same holds for the acronym MAA (Mutually Assured Awareness).

same token, a "Mutually Assured Awareness" (MAA) equilibrium, where everyone acquires information, is also possible. The path taken depends critically on the agents' expectations about the actions of others in their group. Conversely, a group of Substitutes cannot fall into either a MAI or MAA equilibrium, as its agents are inherently inclined to resist such dynamics.[6]

The next step in our investigation is an experiment to explore how strategy types manifest in practice and how beliefs about others' choices influence information avoidance in groups.

3.2 Social Preferences and Information Avoidance in Groups

In this subsection, we extend the basic theoretical framework to allow for the possibility that agents have prosocial preferences.

Altruism First, we consider the possibility that individuals have altruistic preferences. Specifically, given a date-2 material payoff u, common to all group members, an agent i derives a date-2 utility equal to $(1 + \alpha_i)\, u$, where $\alpha_i > 0$ is an altruism parameter.

Lemma 4 *Altruism has an ambiguous effect on information avoidance.*

Intuitively, since agent i cares about the payoff of other members of her group, the altruism parameter acts as an amplifier of the date-2 payoffs. On the one hand, more altruistic individuals have stronger incentives to acquire information since they internalize more the positive returns that this action will generate on the payoffs of other members. On the other hand, a higher altruism parameter spreads out the payoffs across states, increasing the variance of the expected utility lotteries associated with the decision to acquire information. Unless v_i is everywhere convex —in which case the agent always acquires information independently if the value of α_i— this higher variance may increase the agent's incentives to avoid information. In general, therefore, higher altruism produces countervailing forces whose net effect is ambiguous.

Reciprocity To be completed.

[6]Consider, for instance, a group composed of Substitutes with a 0.5 threshold. It is easy to show that, in the unique equilibrium, exactly half the group acquire information and half avoid it.

4 Experimental Design and Procedures

4.1 Experimental Design

We investigate whether, in groups, information avoidance is interdependent. We specifically ask: How do individuals respond, in terms of information avoidance, when they perceive that information avoidance is more prevalent within their group?

Decision environment To study interdependence in information avoidance, we design an experiment in which individuals in a group choose whether to acquire or avoid costless information. After being randomly assigned to groups, participants are told that, at the end of the experimental session, a state of the world (either good or bad) will be randomly drawn. In the bad state (or the “Screams" state), the group will experience a negative consumption event: listening to a series of unpleasant screams. In the good state (or the “Quiet” state), no such event happens.

Each member of the group can choose one of two options. If she chooses option “Now”, she learns the state of the world immediately, while if she chooses option “Later” she defers learning the state of the world until it is revealed in the second part of the experiment.[7] The choice between “Now” and “Later” is the key decision of interest. In this paper, we refer to this binary choice as the “information decision”.[8]

A distinctive feature of our experiment is that information decisions generate a payoff externality: the acquisition of information creates a positive return not only for oneself (reflecting an instrumental value of information), but also for other members of the group (externality). If the state is *Screams*, the volume of the screams that all group members will hear decreases in the total number of group members who acquire information (option Now). More precisely, each individual knows that by choosing Now, she lowers the volume of the screams in the Screams state by 2 points *for all* group members.[9] There are no monetary costs involved with choosing either Now or Later, so that information about the state is readily available.

A key distinction from previous studies on information avoidance in individual decision-making (e.g., Falk and Zimmermann, 2024) is that our study examines decision-making

[7]Participants are aware that if they choose Later, they will be informed about the outcome just before the period in which screams are potentially played.

[8]In each group, decisions are simultaneous. Therefore, there is no social learning among members of the group.

[9]The contribution to the volume of screams is additively separable. Therefore, we examine whether informational decisions can exhibit interdependence even in the absence of in-built interdependencies in technology.

within a group context influenced by externalities—a context that has not been explored before. Accordingly, our analysis begins by examining whether information avoidance occurs at all in our group setting. This is a first, necessary condition to study interdependencies in informational decisions.

Treatments and average interdependence We aim to identify whether information decisions are interdependent. To this end, we designed our experimental treatment to induce exogenous variation in participants' beliefs about the prevalence of information avoidance in their group. We do so through an information provision treatment, where we randomize information about the prevalence of information avoiders in the past. There are two conditions: the *Many Avoided* and *Few Avoided* treatments, randomized at the group level. In *Many Avoided* (*Few Avoided*), we inform that 80% (20%) of subjects in a particular group of a past pilot similar to the current experiment chose to avoid information about the state. Participants are also reminded of the potential impact on scream volume if their group members exhibit similar information avoidance behaviors as those observed in the pilot study.[10] To check whether the treatment manipulation works as intended, after the information provision treatment we elicit the participants' beliefs about the fraction of avoiders among their group mates.

To assess the (average) interdependence of information decisions, we compare the fraction of information avoiders (i.e., people who chose option Later) between *Many Avoided* and *Few Avoided*.

Heterogeneity of interdependence As Proposition 3.1 suggests, individuals may be heterogeneous in how they respond to others' information decisions. To delve into potential heterogeneity in individual responses, we use the strategy method (Selten, 1967) to examine how each participant responds to a range of choices of their group mates, in an incentive compatible way. To do so, we elicit the full schedule of each individual's best response to all the possible choices made by others in the group. We classify the best response schedules based on whether a participant chooses Later when the share of avoiders among other members is sufficiently large (Strategic Complement) or sufficiently small (Strategic Substitute). If the participant's choice is the same for all distributions of information

[10]Specifically, we inform each participant that, if their group mates behave like the participants of the pilot, the volume in the *Screams* state would be, depending on whether she herself avoids or acquires information, 89 or 91 (in the *Many Avoided* condition), and 60 or 62 (in the *Few Avoided* condition). Conditional on her choice, the difference in the severity of the screams across conditions arises from the information choices of others.

avoidance in their group, we classify them as Always Getter (if they always choose to acquire information) or Always Avoider (if they always choose to avoid information). We refer to these categories of best responses as "strategy types", paralleling the theoretical classification defined in Section 3.1. Figure 1 provides examples for each strategy type, as measured in the experiment.

Prosocial preferences In our decision environment, information decisions generate payoff externalities on other members. Therefore, prosociality may be a relevant predictor of preferences for information. Intuitively, given that information decisions modify other members' payoffs, social preferences may predict both unconditional and conditional informational decisions and moderate the treatment effect on information avoidance. To explore how social preferences and other preference traits affect information choices, we elicit measures of social preferences by administering the Global Preferences Survey (GPS) (Falk et al., 2018), which measures altruism and reciprocity, as well as trust, risk attitudes and time preferences.

Mediators of interdependence To shed light on mechanisms and determinants of information decisions, we introduce a novel elicitation designed to measure anticipatory utility over the future consumption event (screams or silence). A crucial feature of our elicitation is its granularity: anticipatory utilities are elicited *conditional on each information set*, namely, upon acquiring information and finding out a positive signal about the state ("good news" event), upon acquiring information and discovering negative signal ("bad news"), and avoiding information and discovering no signal at all ("no news"). Specifically, we ask participants: a) "How happy would you be *thinking about the outcome ahead* if you chose Now and then discovered that the lottery outcome is Screams?"; b) "How happy would you be *thinking about the outcome ahead* if you chose Now and then discovered that the lottery outcome is Quiet?"; and c) "How happy would you be *thinking about the outcome ahead* if you chose Later?" In addition, we elicit the participants' subjective belief that the state is Screams (the prior belief of finding out bad news). This set of four questions mirror the components of expression (2), the net utility of avoiding information (relative to acquiring it), which theoretically drives the information decision. This granular elicitation of the components of the net utility enables us to study how the treatment changes the expected anticipatory utility of, e.g., acquiring information and learning good or bad news separately, which

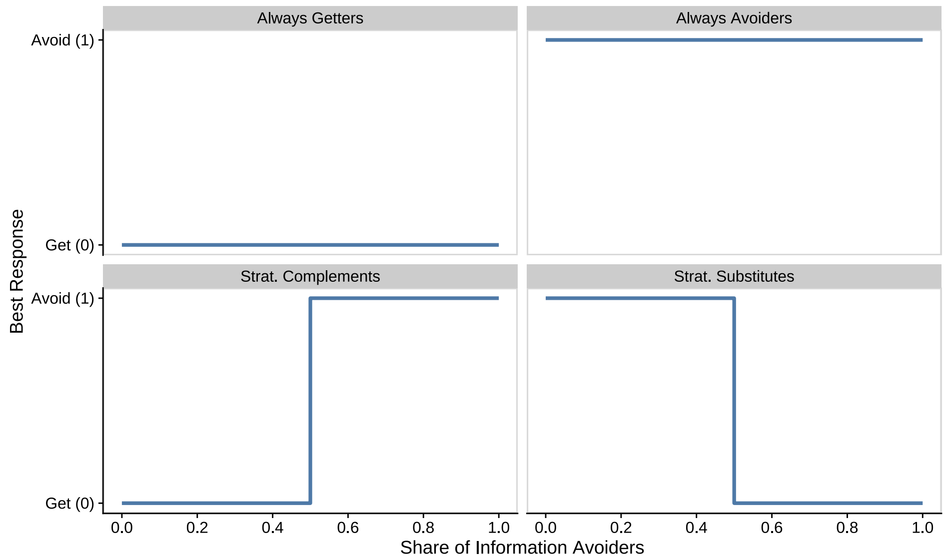


Figure 1: Examples of strategy types

Notes: The figure shows examples of individual best response schedules for each category of best response ("strategy type").

helps to distinguish among mechanisms. We call this set of questions the "Kreps-Porteus questions" (or "KP questions").

Task and earnings Following the design in Falk and Zimmermann (2024), we include an incentivized task for participants to complete in the first part of the experimental session, after the participants have taken the Now or Later decision and before the second part of the experiment where they may or may not experience the screams depending on the realization of the state. The experimental timeline is summarized in Figure 2.

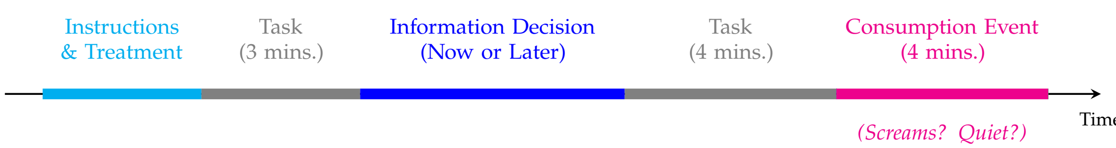


Figure 2: Timeline of experiment

4.2 Experimental Procedures

The experiment was conducted in the Centre for Decision Research and Experimental Economics (CeDEx) Lab at the University of Nottingham. 493 participants across 18 experimental sessions were recruited from the CeDEx subject pool. The average session lasted 1 hour.[11]

At the start of each experimental session, participants were matched into groups, and each group was randomly assigned to a treatment condition.[12] Participants were then provided with detailed instructions on computer screens.[13] The instructions included a description of the state of the world lottery, the negative consumption event (screams), the timeline of the experiment, the informational decisions and its implications, and the task. In order to ensure understanding of the decision environment, we included control questions about the timing of learning the outcome and about the nature of returns and externalities from information acquisition/avoidance. During the instructions phase, we also implemented our information provision treatment.

A computerized lottery determined the outcome (either *Screams* or *Quiet*) for each group. If the outcome was *Screams*, in the second part of the experiment the members of the group would hear a series of unpleasant screams through headphones they were asked to wear during the whole session. We administered the scream sounds used in Beaurenaut et al. (2020).[14] To ensure the hearing safety of participants, the maximum possible volume level in the experiment, level 100, was calibrated at 80dB, in compliance with the UK hearing safety regulations. The calibration was done for each pair of headphone and computer station, ensuring that no subject would be exposed to unsafe levels of hearing. To enforce the take-up of the potential screams, we informed at the beginning of the session that the use of headphones was mandatory during the entire session and that attempts to break this rule would be a basis for expulsion from the session. To ensure compliance with this rule and that participants wore the headphones correctly, we placed two invigilators who

[11]The experiment was preregistered, AsPredicted #153871 and #175588. The experiment was approved by the University of Nottingham's School of Economics Research Ethics Committee. At the start of the session, participants were informed about the nature of the experiment and provided written consent to participate.

[12]The random assignment of the treatment was blocked by week of year and time of session (morning and afternoon).

[13]The program was written in Lioness (Giamattei et al., 2020).

[14]While our participants did not listen to any sample screams before they made their informational decisions, they were informed that "[p]revious research has shown that people consider similar screams as *more aversive (disturbing) than* the sound of *nails sliding on a chalkboard*" Evidence backing this statement can be found in Kumar et al., 2009.

monitored these conditions in each session. Participants were told they could leave the session at any time.[15]

The use of screams instead of electric shocks (employed in experimental studies on information avoidance in individual decision making) is based on three rationales. First, one of the motives that may lead to information avoidance are anticipatory emotions, such as anxiety about a future outcome, and studies show that the screams are able to sustain anxiety for longer periods than electric shocks (Beaurenaut et al., 2020). This feature is especially desirable for experiments with group decision-making, which typically take longer compared to individual decision-making. Second, the provision of human scream sounds is more scalable than electric shocks, since the only required equipment are computers and headphones, as opposed to devices specialized for the provision of shocks.[16] Non-scalability imposes a natural limit to group sizes, which, as discussed below, affect the likelihood of the emergence of complementarities in information decisions (Bénabou, 2013). Third, screams allow a more granular manipulation and more intuitive understanding of variations in severity (i.e., volume levels as opposed to electric shock intensity); this granularity in severity is important as group sizes become larger and the marginal return of acquisition of information by any individual member becomes smaller. Fourth, the use of human screams alleviates ethical concerns compared to electric shocks.[17]

The experimental and model assumption is that screams are a bad. To assess whether this is the case, we elicit the participants' utility over two different levels of volume of screams. We ask: "Imagine that the Outcome is Screams, and that the volume of the screams will be level 100. How happy would you be thinking about the Outcome ahead?"; we ask a similar question for volume level 50. This allows us to identify subjects who, contrary to our assumption, derive higher utility from higher levels of volume of scream sounds ("volume lovers").

For groups whose lottery outcome was *Screams*, the volume of the screams was determined

[15]Only one participant left, after about an hour of the session, reportedly due to the start of a lecture.

[16]The lower implementation barrier also facilitates replicability.

[17]Recently, experimental literature in Economics has started using the prospect of money losses as a negative future event (Pagel, 2018, Engelmann et al., 2024). In particular, Engelmann et al. (2024) finds that money losses induce anticipatory anxiety and wishful thinking about the future event. The literature on intrinsic preferences for information has focused attention on future *consumption* events as distinct from *income* events, because knowing the latter introduces a planning advantage that motivates the acquisition of information about the future income event. To avoid introducing this extra planning advantage, in our experiment, we follow the latter tradition and focus on a consumption event. We argue that, theoretically, the relevant feature of the event is that the prospect of hearing human screams, much like the prospect of money losses, is perceived as a bad event by participants. In Section 5.1, we show evidence that, in our experimental sample, this is the case.

by the information decisions of its members. Specifically, the volume of the screams in group g was determined by the following function:

$$vol_g = 50 + (100 - 50)\left(\frac{1}{N_g}\sum_{i=1}^{N_g} Avoid_{ig}\right), \tag{3}$$

where $vol_g \in [50, 100]$ is the volume of screams for group g conditional on the lottery outcome being Screams, N_g is the number of members in group g, and $Avoid_{ig} = 1$ if subject i in group g chooses Later and 0 otherwise. The acquisition of information about the state thus partially mitigates the volume of the screams.

While the volume production function in Equation (3) is formally reminiscent of a public goods game (PGG), our experimental design differs in several ways that reverse the standard theoretical predictions. First, the nature of the cost is distinct: in a PGG, the cost of contributing is a known monetary amount, identical for all players and independent of others' actions. Here, there is no monetary cost of contributing; the cost is instead psychological (from potentially learning bad news about the future) and is heterogeneous across individuals. Moreover, the psychological cost plausibly varies with the choices of others (which determine how severe the bad state is). These differences lead to a stark divergence in the prediction for a purely material-payoff maximizing agent: such an agent would optimally free-ride in a standard PGG but would always prefer to "contribute" (i.e., acquire information) in our setting, a dominant strategy for maximizing material outcomes.

Each group was composed by an entire experimental session, with group sizes of approximately $N_g = 30$. This implies that the return to each participant's information acquisition (in terms of volume reduction) was $1/30 * 50 = 1.67$ volume points.[18]

The information decision, the binary decision between options Now and Later, is our main outcome of interest. To obtain an alternative, continuous measure of intensity of preferences, we also ask participants, after they chose either Now or Later, to rate the strength of preference for their selected option on a likert scale ranging from 0 ("I am indifferent") to 10 ("I have a very strong preference for the selected option"). From the measure, we construct a continuous version of the information decision, weighted by the reported strength of preference. This continuous outcome variable ranges from -10 ("I have a very strong preference for option Now") to 10 ("I have a very strong preference for option

[18]We described to participants the return to information acquisition as being equal to "around 2 out of 100 volume level points".

Later"). We also ask participants to write down the reason for the selected option in an open text box.

To explore individual heterogeneity in best responses, we use the strategy method (Selten, 1967) to elicit each participant's complete schedule of choices between options Now and Later as a function of others' choices. This schedule captures their choice conditional on every possible share of information avoiders among others in the group, specifically for the shares: 0%, 20%, 40%, 60%, 80%, and 100%.

We make the strategy method elicitation incentive-compatible by assigning a strictly positive probability that these choices are implemented. Specifically, before the elicitation, we inform participants about the following procedure. After the group members make their (unconditional) informational decisions (between Now and Later), we randomly select $N_g - 1$ subjects and calculate the proportion of information avoiders *among these* $N_g - 1$ *subjects*, based on their unconditional decisions. For the remaining participant, we implement her selected best response to this share.

To measure general preferences for information, we administer the "Information Preferences Scale", or "IPS" (Ho et al., 2021). This scale measures an individual's desire to obtain or avoid information that may be unpleasant but is instrumentally valuable, across several domains.[19] The scale ranges from 1 = "Definitely not want to know" to 4 = "Definitely want to know". Therefore, higher IPS scores indicate greater general willingness to obtain information. Measuring general preferences for information allows us to, first, assess the relationship between general preferences for information and the specific information decisions in our experiment; second, to check that the randomization is balanced in this dimension, which is potentially important in determining informational decisions; and third, to control for general preferences for information in our analysis.

To measure social and economic preferences, we administer the Global Preferences Survey (GPS) (Falk et al., 2018). The GPS measures risk attitudes, time preferences, altruism, reciprocity, and trust. Each dimension of preferences is elicited through the responses to two questions. Following the procedure in Falk et al. (2018), we standardize the answers, transforming them into z-scores.[20] Higher z-scores indicate, respectively, higher risk seeking,

[19]The scale is constructed by averaging the answers to 18 questions that ask the subject's willingness to obtain information in hypothetical scenarios. The scenarios involve five domains: health, finance, social relations, ego-relevant characteristics, and occupation. For each question, the 4 possible answers are "Definitely don't want to know" (encoded as 1), "Probably don't want to know" (2), "Probably want to know" (3), and "Definitely want to know" (4).

[20]For risk attitudes and time preference, one of the questions used by Falk et al. (2018) involve the staircase

patience, altruism, reciprocity, and trust. We also elicited demographics (e.g., age, gender), prior work experience, ratings about the perceived difficulty of instructions and quiz questions. Participants were given the opportunity to feedback on their experience in the experiment in open text boxes.

Following Falk and Zimmermann (2024), we include an incentivized task after the Now or Later decision and before the second part of the experiment. Participants are asked 90 general knowledge quiz questions, with earnings increasing in the number of correct answers. Each participant is paid according to her own performance, which is the only source of monetary earnings in our experiment, apart from the show-up fee.

4.2.1 The sample

We conducted 17 experimental sessions (each constituting an independent experimental group) across two data collection waves (December 2023 and May-June 2024). We recruited participants from the subject pool of the Centre for Decision Research and Experimental Economics (CeDEx), at the University of Nottingham. The sample for analysis consisted of 465 participants, with an average of 27.35 ± 3.66 subjects per session.[21] Participants earned on average £12.52 (± 2.00), including a £5 show-up fee. Demographically, participants were on average 22.51 (± 3.71) years old, and the sample showed balance in terms of gender. The average score on the Information Preferences Scale (IPS) was 3.12 ± 0.41, indicating a general preference for obtaining potentially undesirable information across domains.[22] Full descriptive statistics of baseline characteristics are available in Table C.1 in the Appendix.

5 Results

This section presents the analysis of our experimental data. We begin by establishing two necessary preconditions for our study. In Section 5.1, we verify that, for most participants, screams are perceived as a bad, confirming a fundamental theoretical assumption. In Section 5.2, we document the existence of information avoidance in our decision environment

method. Since administering this could be time-consuming and cognitively burdening in the middle of our experiment, we did not include such questions.

[21]Due to technical issues, data (including the main outcomes of interest) was not recorded for 22 subjects in a single session of 28 participants and is excluded from the analysis; our results are robust to the inclusion of the 6 individuals from this session for which data is available. In a separate incident, one participant left early, about an hour after the session started, reducing observations for final-stage variables by one.

[22]As a reminder, a score of 3 means "Probably want to know".

(a context where the positive instrumental value and externalities from acquisition of information create an incentive to *acquire* it). We further establish that the decision to avoid information in our environment is likely not random nor driven by indifference, as avoiders report strong preferences for their choice.

Following these prerequisite validations, we turn to our primary results. In Section 5.3, we leverage experimental variation in beliefs about others' information decisions to test for strategic interdependence, assessing whether the likelihood of information avoidance increases (strategic complementarity) or decreases (strategic substitutability) on average. In Section 5.4, we move beyond the average to characterize individual heterogeneity. Using the strategy method, we document the empirical distribution of strategy types and report heterogeneous treatment effects by type. In Section 5.5, we explore the mechanisms driving the average treatment effects. Specifically, we use the KP questions to decompose the effect by identifying which components of the expression for the net incentive to avoid information (2) are most impacted by the treatment. We also discuss the plausibility of alternative mechanisms. Finally, the next section is devoted to analyzing how the group composition, in terms of strategy types, shapes the distribution of equilibrium avoidance.

5.1 Are screams a bad?

Before presenting our primary findings, we first conduct a validation check of a foundational assumption underlying our study: that participants actually perceive the prospect of listening to screams as a bad. We provide evidence confirming that the assumption holds for our sample. First, we asked subjects how "(un)happy" they would be if screams were played at volume 50 and, separately, at volume 100. The difference in these ratings provides a proxy of each participant's disutility of listening to screams at higher volume. Comparing these ratings reveals that a large majority of subjects report lower utility levels for screams at higher volumes: specifically, 82% of subjects report that screams at volume 100 would be strictly worse than at volume 50.[23]. Second, using the KP questions, we asked subjects to rate their "(un)happiness" upon discovering that the state was Screams after choosing Now, and separately, if it was Quiet. The difference in these ratings provides a measure of each participant's utility for listening to screams relative to silence. If the screams are perceived as a bad, we would expect lower ratings for the Screams state than for the Quiet state. As

[23]Among the remaining subjects, 3% are indifferent between volume levels, and 9% would prefer volume 100. We call the later type of subjects "volume-lovers".

expected, this pattern holds for the majority of subjects: 81% report that discovering that the Screams state is strictly worse than discovering that the Quiet state.[24] Taken together, both pieces of evidence support the view that the vast majority of subjects perceive screams as a bad.

Furthermore, the fact that higher-volume screams are perceived as worse than lower-volume ones is compatible with the notion that participants perceive the instrumental value of information acquisition (which allows them to reduce the volume of screams) as non-trivial.

For our main analysis, we include all subjects, including those with weak preferences for screams and volume (or both). We refer to such subjects respectively as "scream-lovers" and "volume-lovers".[25] Analyses that exclude these individuals are presented in the Appendix Section B.1. They confirm that our findings are robust to the exclusion of volume- and scream-lovers, and, in some cases, become even stronger.

5.2 The prevalence of information avoidance in groups

To assess the occurrence of information avoidance, we focus on the information decision —the binary choice between learning the state Now or Later. We code the decision as $Avoid_{ig} = 1$ if individual i in group g chose Later, and 0 if she chose Now. We also construct a continuous measure, the "information avoidance scale" (or "IA scale"), to capture the *strength of preference* for either option.[26] Analyses of both measures yield very similar qualitative results. Therefore, we report findings for the binary information decision here, and defer the results for the IA scale to the Appendix Section E.

We find statistically significant evidence of information avoidance. Panel (a) of Figure 3 shows the incidence of information avoidance both by data collection wave and in the pooled sample. The incidence of information avoidance in the pooled sample is 0.286. This is significantly different from zero at the 1% level. The incidence is robust across data collection

[24]Among the remaining subjects, 6% are indifferent between the two states and 14% would prefer discovering the Screams state. We call the later type of subjects "scream-lovers". The distribution of types of preference screams and volume is shown in Figure B.2 and Figure B.1 in the Appendix.

[25]There are 111 subjects who are either scream lovers or volume lovers or both. The distribution of preferences is reported in Appendix Section B.

[26]This scale is constructed as follows. After participants made their information decisions, we asked them to rate the strength of their preference for the selected option (from 0="I am indifferent" to 10 = "I have a very strong preference"). For a subject who chose Later, her IA scale is equal to her stated strength of preference. For a subject who chose Now, her IA scale is equal to the negative of her stated preference strength. Thus, the IA scale ranges from -10 (very strong preference for Now) through 0 (indifference between Now and Later) to 10 (very strong preference for Later), providing more variation than the binary measure.

waves.

This result relates to other findings in the literature. In an individual decision making context, Falk and Zimmermann (2024) find that 52% of subjects avoid information. The difference from our finding could be attributed to a series of factors. In addition to some differences in implementation, their setup implies that information provides no material benefit (i.e., it does not affect the severity of the future negative consumption event). This removes the instrumental motivation for information acquisition, making a higher degree of information avoidance more likely. Prosocial motives for information acquisition are accordingly also absent.

The decision to avoid information in our sample does not seem to be driven by indifference nor subject to trembling. Panel (b) of Figure 3 shows the distribution of preference strength for avoiding information, in the subsample of subjects who chose to avoid information. The preference strength was elicited on a scale ranging from 0 ("I am indifferent") to 10 ("I have a very strong preference"). The average strength of preference is 6.6, far from indifference. Moreover, the cumulative distribution (depicted by the orange line) shows that only 6% of information avoiders report being indifferent between Now and Later, and more than 80% report a preference strength of at least 5. This shows that for the majority of avoiders the informational decision is not driven by indifference, but is rather strongly preferred.

The decision to avoid information is predicted by individual characteristics. In the Appendix, Figure C.1, we report the results of a regression of information avoidance on individual characteristics: economic and social preferences, information preferences, demographics, and field of study. As expected, the IPS scale predicts the decision to avoid information in our experiment. In the economics and social preference domain, only altruism predicts information avoidance. As discussed in Lemma 4, the theoretical prediction on the effect of altruism on information avoidance is ambiguous. Our data shows that empirically, in our sample, altruism is significantly associated with a lower likelihood of avoiding information ($\hat{\beta} = -0.068$). The finding is consistent with the idea that prosocial individuals are more likely to acquire information to benefit others.

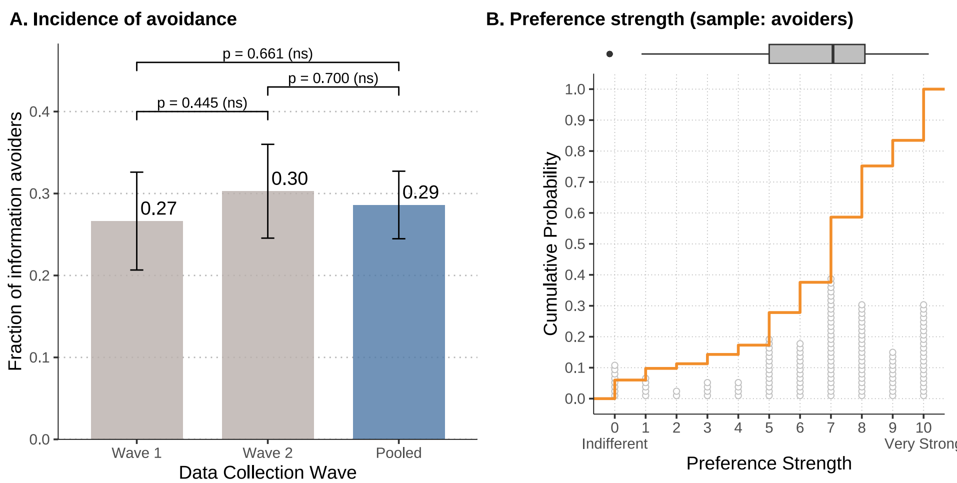


Figure 3: Information Avoidance: Incidence and Preference Strength

Notes: **Panel A** shows the incidence of information avoidance. The bars show the proportion of subjects who chose to avoid information (option Later). The first two bars (brown) show the proportion by data collection wave, while the third bar (blue) shows the proportion in the whole sample. The whiskers show 95% confidence intervals; p-values from two-sample proportion tests. **Panel B** shows the distribution of the strength of preference for avoiding information in the subset of subjects who chose to avoid information. The preference strength is elicited on a Likert scale ranging from 0 ("I am indifferent" between Now and Later) to 10 ("I have a very strong preference" for avoiding information). The frequency of preference strengths is shown by the gray dots, with each dot representing a subject. The cumulative distribution is shown by the orange line. The boxplot shows the median and the interquartile range of preference strength, with whiskers representing the most distant point within 1.5 times the interquartile range.

5.3 The interdependence of information avoidance: evidence from the experiment

We now turn to the question of whether in groups information avoidance decisions are interdependent, starting from the results of the information provision treatments.

Treatment balance checks are reported in the Appendix Table C.2. All individual characteristics are balanced across conditions, with the exception of the representation of some fields of study.[27] We control for the field of study in the regressions. Importantly, the IPS scores are balanced across treatment groups, for both the overall scale and for each of the domain-specific subscales. Therefore, any difference in information avoidance between treatments is not attributable to baseline differences in general or domain-specific preferences for information.

Our treatment manipulation is effective in shifting expectations about information decisions by other members of the group. Panel (a) in Figure 4 shows the average belief by condition. As intended, the information provision treatment substantially shifts beliefs upwards. The difference in average beliefs about peers' informational decisions is large and significant: on average, subjects in *Many Avoided* believe that 0.219 higher proportion of their peers would avoid information, compared to those in the *Few Avoided* condition. The difference represents a 59.12 % increase relative to the Few condition, and is statistically significant at 1% level. Moreover, the shift in beliefs is not only on average: Panel (b) displays the empirical CDFs of beliefs by condition, showing that the distribution of beliefs in *Many Avoided* first-order stochastically dominates (with a small number of exceptions at the extremes of the distribution) the distribution in the *Few Avoided* condition. In sum, our experimental treatment, designed to shift beliefs about the behavior of other members, seems to have been successful.

Table 1 reports a linear regression of the belief on the treatment dummy. On average, the expected share of avoiders among other group members is 21.9 percentage points higher in Many than in Few, representing a 59.1% increase. The difference is highly significant and robust to the inclusion of demographic controls, measures of economic and informational preferences, and strata fixed effects. Taken together, the results suggest that the treatment effectively manipulates beliefs about the share of avoiders among other members of the group.

[27] Compared to Few, there is a lower share of both Business and Economics students in Many.

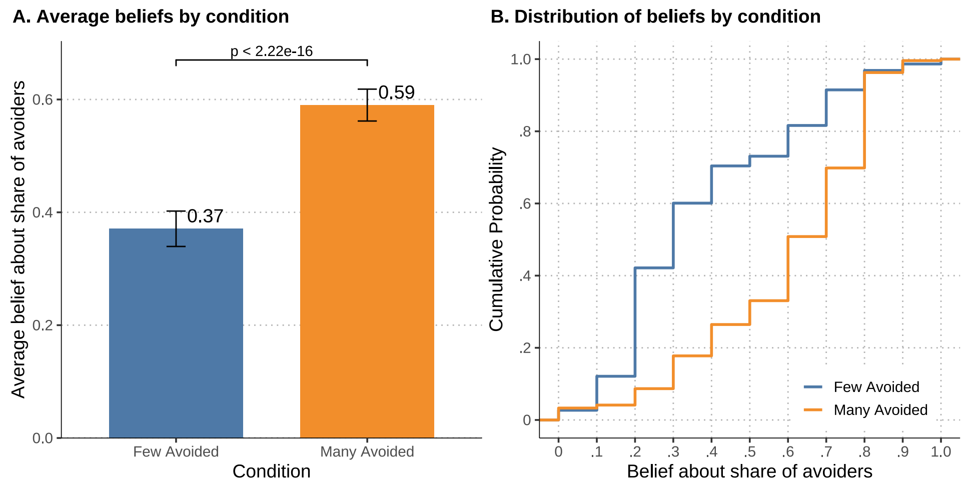


Figure 4: Effect of provision of information about past information choices on beliefs about information choices among other members of group

Notes: The figure shows the distribution of participants' beliefs about the share of information avoiders among other members of their own group, by condition. Panel (a) shows the average belief by condition. The error bars show 95% confidence intervals. Panel (b) shows the empirical cumulative distribution of beliefs by condition.

	Dep. Var: Belief about share of avoiders among others			
	(1)	(2)	(3)	(4)
Many Avoided	0.219***	0.215***	0.218***	0.221***
	(0.0215)	(0.0252)	(0.0254)	(0.0253)
Constant	0.371***	0.471***	0.394***	0.494***
	(0.0160)	(0.0720)	(0.137)	(0.163)
Strata FEs		✓	✓	✓
Demographics			✓	✓
Field FEs			✓	✓
Economic preferences				✓
Information Preferences Scale				✓
Observations	465	465	460	460

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table shows the estimated coefficients of a regression of beliefs about the share of information avoiders among other members on the treatment dummy. The dependent variable is a continuous measure of beliefs, ranging from 0 ("no one will avoid information") to 1 ("everyone will avoid information"). The treatment dummy is equal to one if the subject was in the Many treatment and zero if in the Few treatment. Demographic controls include age and gender. Field of study fixed effects include dummies for fields of study. Economic preferences include risk-seeking, time preferences, altruism, reciprocity, and trust. The Information Preference Scale measures an individual's general preference for obtaining information. Strata-fixed effects include dummies for week of year and time of day.

Table 1: Treatment effect on belief about share of avoiders among other members

Since beliefs about other members' information decisions exogenously increase in the *Many Avoided* condition relative to the *Few Avoided* condition, do we observe any difference in the actual fraction of avoiders across conditions? In other words, is there any interdependence of information decisions, and how strong is it? We assess the hypothesis of whether the proportion of subjects who choose to avoid information differs between Many and Few. Figure 5 shows the proportion of information avoiders across treatment conditions. Compared to Few, the share of information avoiders in Many is 0.084 higher ($p = 0.045$). This difference represents a 34.81% increase relative to the Few condition.

Table 2 reports the coefficients from a regression of the information avoidance dummy on the treatment and covariates, in a linear probability model. The unconditional treatment effect (column 1) is 0.084, significant at 5%. This estimate is robust to the inclusion of individual characteristics and strata-fixed effects (0.099, significant at 5%). When all controls are included, the treatment effect increases to 0.123, becoming significant at 1%. In this specification, the treatment effect represents half of the baseline proportion of avoiders.

This estimate of the treatment effect is robust to a variety of alternative specifications. We find even stronger treatment effects when scream- and volume-lovers are excluded from the sample (Tables B.2 and B.3 in Section B.1). We also identify qualitatively similar results when the outcome variable is the continuous preference strength for information avoidance (Appendix Section E). Finally, the estimate is robust to the inclusion of different subsets of controls (Appendix Section D) selected by the post double selection procedure (Belloni et al., 2014).

Social preferences are predictive of the information decision and moderate the interdependence. In particular, reciprocity interacts with the treatment. Since avoiding information is a way of harming others and taking revenge, more reciprocal individuals are expected to be less inclined to avoid information in the *Few Avoided* condition (where the majority is expected to acquire information and hence do "good" by improving the outcome) than in the *Many Avoided* condition (where the majority is expected to avoid information and harm by worsening the outcome). In Panel B of Figure 5, we include in the regression interactions of the treatment with social preferences and find that, indeed, more positively reciprocal subjects are less likely to avoid information in the *Few Avoided* condition, but are more likely to avoid information in the *Many Avoided* condition, and the interaction is significant at 10%.

In summary, our information provision experiment provides robust evidence that, relative to Few, in Many subjects both believe that a higher share of their group members are

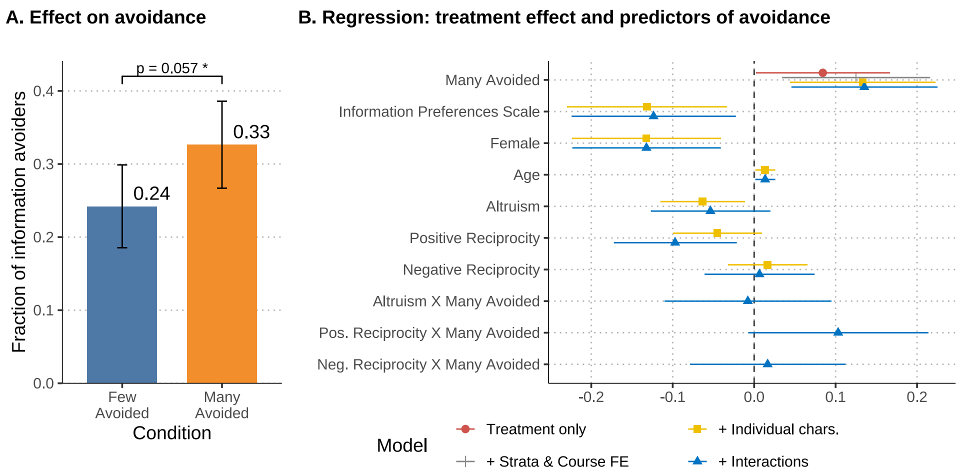


Figure 5: Proportion of avoiders: treatment effect and predictors

Notes: **Panel A** displays the fraction of information avoiders, separately for the *Few Avoided* condition (where subjects were told that 20% of participants in a particular group of a prior similar experiment avoided information) and the *Many Avoided* condition (where the analogous fraction was 80%). Error bars indicate 95% confidence intervals. p-values are derived from a two-sample proportion test.
Panel B presents coefficients from a linear regression of the information decision Avoid_{ig} (equal to 1 if individual i in group g chose to avoid information, and 0 otherwise) on the treatment dummy and additional covariates. The first model (red dots) includes only the treatment dummy. The second model (gray plus signs) adds strata and field of study fixed effects. The third model (yellow squares) further controls for individual characteristics: the Information Preferences Scale (Ho et al., 2021; with higher values indicating a stronger overall preference for *knowing* potentially undesirable information), gender, age, social preferences (altruism, positive and negative reciprocity), and economic preferences (risk aversion, patience, and trust; omitted from the figure for conciseness). The fourth model (blue triangles) incorporates treatment interactions with social preferences. Error bars represent 95% confidence intervals computed based on robust standard errors.

	Dependent Variable: Avoided Information					
	(1)	(2)	(3)	(4)	(5)	(6)
Many Avoided	0.0843**	0.0994**	0.112**	0.122**	0.123**	0.124**
	(0.0418)	(0.0489)	(0.0489)	(0.0491)	(0.0487)	(0.0488)
Age			0.00966	0.0123*	0.0144**	0.0143**
			(0.00626)	(0.00630)	(0.00633)	(0.00641)
Female			−0.136***	−0.132***	−0.127***	−0.127***
			(0.0472)	(0.0483)	(0.0484)	(0.0485)
Pos. Reciprocity				−0.0430	−0.0458	−0.101**
				(0.0292)	(0.0289)	(0.0410)
Neg. Reciprocity				0.00542	0.0130	−0.00469
				(0.0258)	(0.0258)	(0.0360)
Altruism				−0.0646**	−0.0651**	−0.0523
				(0.0278)	(0.0278)	(0.0408)
IPS score					−0.130**	−0.118**
					(0.0522)	(0.0537)
Many Avoided X Pos. Reciprocity						0.109*
						(0.0601)
Many Avoided X Neg. Reciprocity						0.0298
						(0.0513)
Many Avoided X Altruism						−0.00642
						(0.0566)
Constant	0.242***	0.321**	−0.0192	−0.0850	0.280	0.273
	(0.0288)	(0.155)	(0.273)	(0.276)	(0.311)	(0.321)
Strata FEs		✓	✓	✓	✓	✓
Field of study FEs			✓	✓	✓	✓
Observations	465	465	460	460	460	460

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table shows the estimated coefficients regression of information avoidance on the treatment dummy in a linear probability model. The dependent variable is a binary indicator equal to one if the subject chose to avoid information and zero otherwise. The treatment dummy is equal to one if the subject was in the *ManyAvoided* condition and zero if in the *FewAvoided* condition. The regressions include strata fixed effects, demographic controls, field of study, and measures of economic and information preferences.

Table 2: Regression of information avoidance on treatment, social preferences, and information preferences

avoiding information, and are more likely to avoid information themselves. This supports the hypothesis that, on average, information avoidance exhibits strategic complementarity, in line with notion of contagious ignorance.

5.4 Heterogeneity in reactions to the choices of others

Section 5.3 showed that exogenously increasing expectations about the share of avoiders in a group raises the likelihood of information avoidance, on average. A key question, however, is whether individual responses to the information decisions of others are homogeneous. As our model in Section 3.1 shows, theory allows for a variety of best-response types. To empirically map this heterogeneity, we employ the strategy method to elicit a *schedule* of best responses conditional on different shares of avoiders in the group. As detailed in Section 4.1, we categorize these empirical schedules based on whether they are increasing in the share of avoiders (Strategic Complements), decreasing (Strategic Substitutes), or constant (either Always Getters or Always Avoiders). This allows addressing an empirical question: do we observe the theoretical heterogeneity in practice? How large is this heterogeneity?

The data reveal substantial heterogeneity in how individuals respond to the information avoidance decisions of others. Panel A of Figure 6 presents the empirical distribution of individual strategy types. The most prevalent type is the Always Getter (42.4%), who seeks information regardless of the share of avoiding peers. At the opposite extreme, the least common type is the Always Avoider (8.8%), who avoids learning the state irrespective of the share of avoiders.

Among the types that exhibit dependence on others' behavior (approximately 40% of our sample), we identify two patterns. 20.6% of participants are Strategic Complements: they avoid information only if a sufficiently high share of their group members also avoids it. The proportion of Strategic Complements is statistically different from zero, with a 99% confidence interval of (0.171, 0.246). A similar share – 19.4% – are Strategic Substitutes, who avoid information only if a sufficiently *low* share of their group members avoids it. The remaining minority (8.8%) are subjects who switch their information decisions multiple times and, hence, whose best responses oscillate. In sum, we find substantial heterogeneity in strategy types, with a significant share of individuals exhibiting dependence on the choices of others.

We now assess whether the treatment effects are heterogeneous across strategy types, examining both the extensive margin (whether the treatment affects the distribution of types)

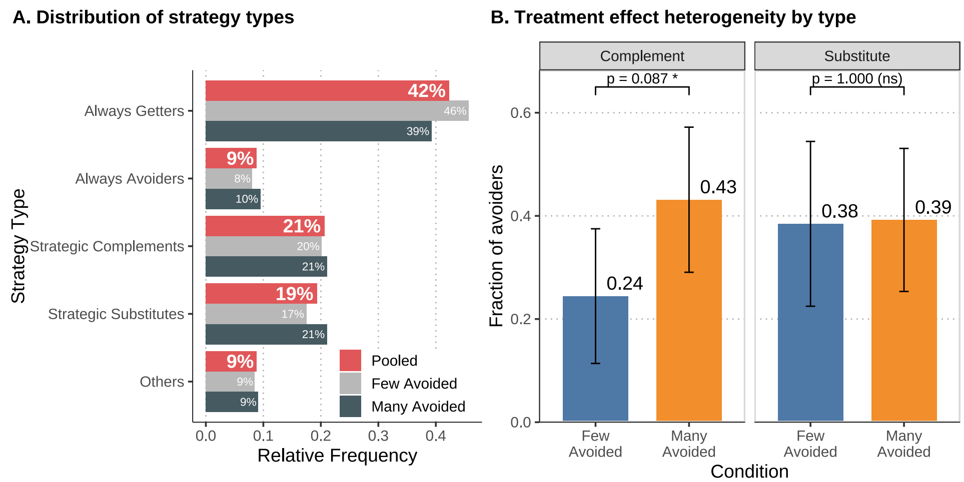


Figure 6: Strategy types: distribution and treatment effects

Notes: The figure shows the heterogeneity of strategy types. **Panel A**: the bars show the relative frequency of strategy types, for each treatment condition (gray bars) and for the pooled sample (red bars). **Panel B** shows the treatment effects on the fraction of information avoiders for the conditional types (Complements and Substitutes). The y axis represents the fraction of subjects who chose to avoid information. Error bars represent 95% confidence intervals; p-values calculated from a two-sample proportion test.

and the intensive margin (the probability of avoidance conditional on the type).

On the extensive margin, we find, reassuringly, that the treatment has no statistically significant effect on the distribution of types. Panel A of Figure 6 shows the distribution split by treatment condition (gray bars), and Table 3 reports the results of regressions where a dummy for each type is regressed on the treatment indicator. The treatment coefficient captures the change in the proportion of a type between *Many Avoided* and *Few Avoided*. We do not find statistically significant differences for any type.

The distributional invariance across treatments is consistent with theory. Best reaction *schedules* themselves should not change unless some (perceived) parameter of the environment, beyond the expected behavior of others, changes. Since the treatment was designed to change only the expectation about others' behavior, but no other feature of the environment, there is no theoretical reason to expect the distribution of types to shift across treatments. Accordingly, we find that, empirically, the treatment has no effect on the extensive margin.

We now turn to the intensive margin. Panel B of Figure 6 compares the avoidance rates for each strategy type across treatment conditions. Table 4 reports a linear regression of the avoidance dummy on the treatment for each type. The treatment coefficient captures the

	(1) Always Getters	(2) Always Avoiders	(3) Complements	(4) Substitutes
Many Avoided	-0.0648	0.0143	0.00895	0.0359
	(0.0460)	(0.0263)	(0.0377)	(0.0367)
Constant	0.457^{***}	0.0807^{***}	0.202^{***}	0.175^{***}
	(0.0335)	(0.0183)	(0.0270)	(0.0256)
Observations	465	465	465	465

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table examines whether the treatment impacts the distribution of strategy types by reporting estimated coefficients from linear regressions of a type dummy on the treatment dummy. Each column corresponds to a different strategy type, where the dependent variable is an indicator equal to 1 if the subject's best response schedule matches that type and 0 otherwise. The independent variable is a treatment dummy equal to 1 for the *Many Avoided* condition and 0 for the *Few Avoided* condition.

Table 3: Treatment effect on distribution of types (extensive margin)

difference in the avoidance rate between the *Many Avoided* and *Few Avoided* conditions for each type. We find that the *Many Avoided* treatment significantly increases the likelihood of information avoidance for Strategic Complements. This is theoretically reasonable: as already presented, the treatment successfully increased subjects' belief about others' avoidance, the argument of best response schedules. For Complements, such schedules are increasing. Therefore, Complements should react to the shift in beliefs by avoiding more frequently. Contrastingly, however, we found no significant response from Strategic Substitutes.

The differential responsiveness can be explained by the interaction between the distribution of switching thresholds and the treatment-induced shift in beliefs, shown in Figure 7. Although the treatment leaves the average switching thresholds unchanged for both types (around 0.5 for Complements and around 0.4 for Substitutes), it substantially increases average beliefs (from 0.35 to 0.62 for Complements and from 0.42 to 0.64 for Substitutes). For Strategic Complements, the belief shift crosses the threshold, triggering increased avoidance. For Strategic Substitutes, however, the new beliefs remain on the same side of the threshold, resulting in no significant change in average avoidance rates.

Finally, columns (1) and (2) of Table 4 show that, as expected, the treatment does not affect the avoidance rate for Always Getters and Always Avoiders, who, by definition, follow strategies that are irresponsive to the behavior of others. Taken together, the pattern of intensive margins of treatment effects by types suggest that the positive average treatment

	Dependent variable: Avoided Information			
	(1) Always Getters	(2) Always Avoiders	(3) Complements	(4) Substitutes
Many Avoided	0.0141	0.0556	0.187*	0.00754
	(0.0332)	(0.0572)	(0.0964)	(0.106)
Constant	0.0490**	0.944***	0.244***	0.385***
	(0.0216)	(0.0572)	(0.0655)	(0.0800)
Observations	197	41	96	90

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table presents estimates from linear probability models where the dependent variable is an indicator equal to 1 if the subject avoided information. Each column reports a separate regression estimated on the sample of a distinct strategy type. The independent variable is a treatment dummy equal to 1 for the *Many Avoided* condition and 0 for the *Few Avoided* condition.

Table 4: Treatment effect on fraction of avoiders, by strategy type (intensive margin)

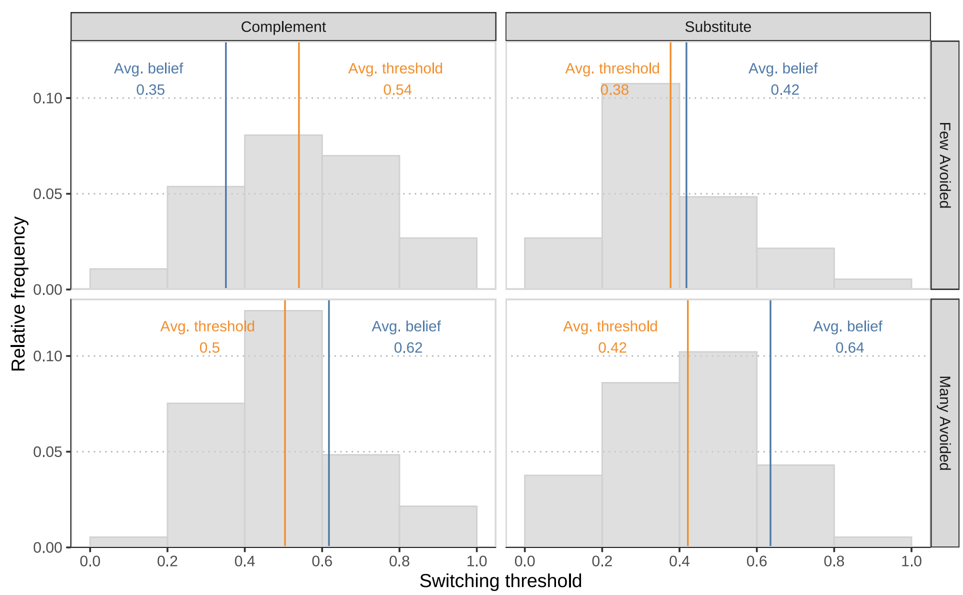


Figure 7: Switching thresholds and beliefs by strategy type and treatment condition

Notes: The figure shows the switching thresholds (i.e., the share of avoiders at which subjects change their choice between avoiding and acquiring information) and beliefs about the share of avoiders, by strategy type and treatment condition. It shows the distribution (gray bars) and the average (orange line) of switching thresholds, as well as the average belief about the share of avoiders (blue line).

effect documented in Section 5.3 is primarily driven by the responsive behavior of Strategic Complements.

Our analysis reveals considerable heterogeneity underlying the average treatment effect. The significant increase in information avoidance is not uniform, but is instead almost entirely driven by Strategic Complements, while the behavior of Always Getters, Always Avoiders, and Strategic Substitutes is unchanged. This pattern of responses is explained by the interaction between switching thresholds and beliefs.[28] Our findings suggest that understanding *which* individuals respond and *how* they respond is important for predicting aggregate acquisition of information. We explore this point further through simulations that vary the composition of the group in terms of strategy types and trace the equilibrium prevalence of information avoidance in Section 6.

5.5 Mechanisms

Section 5.3 showed a positive average treatment effect. In this subsection we explore underlying mechanisms: what makes subjects in the *Many Avoided* condition avoid discovering the state more often than subjects in the *Few Avoided* condition?

The strategic complementarity in information avoidance could be driven by a variety of mechanisms. First, we focus on the "groupthink" channel, formalized by Bénabou (2013) and central in our model: the idea that when a subject expects others to remain ignorant about the state, her own payoffs in the bad state become worse, making it more terrifying to discover bad news about the state. We provide evidence supporting this mechanism in Section 5.5.1 by using our KP measures of expected anticipatory utilities conditional on finding out good, bad, and no news. We then discuss the plausibility of alternative mechanisms in Section 5.5.2.

5.5.1 Anticipatory Utility and Information Decision

To shed light on determinants of information decisions and on mediators of the treatment effect, we use the KP questions to measure the expected anticipatory utility associated to the decisions of acquiring and avoiding information. First, we show that our measures of anticipatory utility significantly predict actual information decisions in our experiment,

[28]Although Strategic Substitutes were unresponsive in this specific intervention, their behavior can potentially change under larger changes in beliefs. The data from the strategy method, which maps the best-response schedule for a wider range of beliefs, allow predicting decisions under such counterfactual beliefs.

above and beyond the predicting capacity of the IPS, an established measure of information preferences. Next, we provide suggestive evidence that the treatment operates mainly through the utilities as captured by the KP measures. Finally, and importantly, we show that the treatment increases the net utility of avoiding information by lowering the expected anticipatory utility of finding out *bad* news, but has no effect on the expected anticipatory utility of finding out *good* nor *no* news, in line with the "groupthink" mechanism in Bénabou (2013) and in our model.

Inspired by our theory section, our KP questions elicit subjects' estimation of their own anticipatory utility *conditional on each possible information set* —that is, upon finding out good news ($\hat{v}_i(g)$), bad news ($\hat{v}_i(b+\Delta)$), and no news ($\hat{v}_i(\hat{q}_i b + (1-\hat{q}_i)g)$)—. In addition, the KP questions elicit the subjective belief that the state is bad ($\hat{q}_i$). All the KP questions are elicited before subjects make their information decisions, so that we interpret them as *expected* anticipatory utilities, where expectations are taken before the subject potentially learns any information.[29]

Based on the answers to the KP questions, for each subject we infer the expected anticipatory utility of acquiring information, by taking the expectation over states

$$U_{0,i}^{Get} \equiv \hat{q}_i \cdot \hat{v}_i(b+\Delta) + (1-\hat{q}_i) \cdot \hat{v}_i(g) \tag{4}$$

based on the reported subjective belief about the state, $\hat{q}_i$. On the other hand, the expected anticipatory utility of avoiding information is simply

$$U_{0,i}^{Avoid} \equiv \hat{v}_i(\hat{q}_i b + (1-\hat{q}_i)g) \tag{5}$$

We then estimate the net expected anticipatory utility of avoiding information φ_i, defined in Equation (2), as

$$\hat{\varphi}_i \equiv U_{0,i}^{Avoid} - U_{0,i}^{Get}, \tag{6}$$

The "hat" notation in $\hat{\varphi}_i$ emphasizes that this is our empirical estimate of the theoretical construct φ_i.

Our model predicts that an individual i should avoid information whenever $\varphi_i > 0$, acquire it if $\varphi_i < 0$, and be indifferent if $\varphi_i = 0$. To assess the predictive power of our

[29] [DISCUSS WHY AND TRADE-OFFS]

empirical anticipatory utility measures, we regress the avoidance dummy $Avoid_{ig}$ on the net expected anticipatory utility, $\hat{\varphi}_i$, and a dummy variable indicating $\hat{\varphi}_i > 0$.[30] To benchmark our measures against an established alternative, we also include the IPS in the regressions. All specifications control for field of study and strata fixed effects, demographic characteristics, and economic preferences.

Results are reported in Panel A of Table 5. Column (1) shows the benchmark. The Information Preferences Scale (IPS) is a significant negative predictor of avoidance, consistent with the interpretation that individuals with a general willingness to *know* potentially undesirable information are *less* likely to avoid information in our experiment. The IPS (together with the mentioned controls) explains only 7% of the variation of choices. Columns (2) and (3) replace the IPS with our measures of anticipatory utility. Both versions of expected anticipatory utility, continuous and binary, are highly significant (1% level) and positively associated to information choices: a higher net anticipatory utility of avoidance is associated with a higher probability of avoiding information. Notably, models that include our measures explain approximately 20% of the variation, nearly three times than that with the IPS. This is possibly because the KP questions elicit aspects that are specific to the decision environment. Columns (4) to (5) include both the IPS and our utility measures. Perhaps surprisingly, the coefficients for both remain remarkably stable in sign, magnitude, with our measures explaining variance above and beyond the IPS. An interpretation is that the information decisions are driven by both a general disposition (captured by the IPS) and environment-specific aspects (captured by the KP measures). This pattern supports both the construct validity and discriminant validity of our KP measures: these are not merely proxies of general preferences for information, but instead elicit aspects of preferences for information that are specific to our environment and predictive of decisions. Overall, our net utility measure is a more powerful predictor of information avoidance, and its predictive power is independent of that of the IPS.

Finally, we assess whether anticipatory utilities mediate the treatment effect in Panel B, where we add the treatment dummy to the regressions. Column (1) includes the treatment dummy but not the measures of anticipatory utility, reproducing the main result of Table 2. Column (2) adds the IPS as a benchmark. The treatment coefficient remains virtually unchanged. Columns (3) and (4) replace the IPS by our measures of net expected

[30] The dummy is equal to 0 if $\hat{\varphi}_i < 0$, and encoded as a missing value for border cases where $\hat{\varphi}_i = 0$ (12 individuals in our sample).

	Dependent variable: Avoided information				
	(1)	(2)	(3)	(4)	(5)
Panel A: Predictive power					
Information Preferences Scale (IPS)	-0.129**			-0.136***	-0.148***
	(0.052)			(0.047)	(0.049)
Net utility of avoiding ($\hat{\varphi}^i$)		0.023***		0.023***	
		(0.003)		(0.003)	
Net utility of avoiding > 0			0.372***		0.379***
			(0.046)		(0.045)
Controls and fixed effects	✓	✓	✓	✓	✓
Observations	460	460	448	460	448
Adjusted R^2	0.071	0.185	0.211	0.199	0.226
Panel B: Treatment and utility					
Many Avoided	0.122**	0.123**	0.061	0.081*	0.062
	(0.049)	(0.049)	(0.047)	(0.045)	(0.046)
Information Preferences Scale (IPS)		-0.130**			-0.136***
		(0.052)			(0.047)
Net utility of avoiding ($\hat{\varphi}^i$)			0.022***		0.022***
			(0.003)		(0.003)
Net utility of avoiding > 0				0.365***	
				(0.047)	
Controls and fixed effects	✓	✓	✓	✓	✓
Observations	460	460	460	448	460
Adjusted R^2	0.070	0.082	0.187	0.214	0.200

Notes: The table shows OLS estimates analyzing the predictive power of the KP measures on actual information decisions. The dependent variable, $Avoid_{ig}$, is an indicator equal to 1 if the subject chose to avoid information about the state and 0 otherwise. "Net utility of avoiding" is the empirical estimate $\hat{\varphi}^i$ of the net expected utility of avoiding information (relative to acquiring it), derived from the answers to the KP questions. "Net utility of avoiding > 0" is an indicator equal to 1 if the empirical net utility of avoiding information $\hat{\varphi}^i$ is higher than zero. "Information Preferences Scale (IPS)" is the scale developed by Ho et al. (2021) measuring overall preference to know potentially undesirable information, with higher values indicating stronger preference to know such information. "Many Avoided" is the treatment dummy, equal to 1 if the individual received the "Many Avoided" treatment and 0 otherwise. Regressions control for demographic characteristics (age and gender), economic and social preferences (patience, risk-seeking, trust, altruism, and positive and negative reciprocity), and fixed effects by strata and field of study of the participant. Robust standard errors in parentheses. Significance levels: $^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: Predictive power of anticipatory utilities (KP measures) on information decisions.

anticipatory utility. The results show a clear pattern: when these utility measures are included, the treatment effect shrinks substantially in magnitude and becomes statistically insignificant (for the continuous utility specification) or less significant (for the binary utility specification). Moreover, comparing the coefficients of the net utility variables in columns (2) and (3) of Panel A (which exclude the treatment dummy) to those in columns (3) and (4) of Panel B shows that including the treatment dummy has no effect on those coefficients. This pattern of findings is consistent with a mediating role for anticipatory utility: the treatment appears to operate primarily by influencing this utility. This theoretically intuitive: the decision should depend only on the associated utility, so once we control for the utility, variations in the treatment dummy should become insignificant. To further illustrate this point, in Figure 8 we report a probit model where the latent variable is the net utility $\hat{\varphi}_i$. The model is estimated separately for the treated (*Many Avoided*) and control (*Few Avoided*) subsamples. The probability of avoiding information increases with the empirical net utility $\hat{\varphi}_i$, and, crucially, this relationship is statistically indistinguishable between treatment conditions. We repeat the same exercise across interdependent strategy types (complements and substitutes, where there is variation in choices) and find the same pattern. This reinforces that our KP questions, designed to measure anticipatory utility, proxy relevant determinants of information decisions and are a key mediating variable of the treatment effect.

Others' information avoidance, anticipatory utility, and the groupthink mechanism Having established the validity of our KP measures, we turn to look deeper at specific components of the net utility of avoiding information. This allows us to provide a decomposition of the treatment effect on such utility, useful to assess the plausibility of the groupthink mechanism.

In separate regressions, we estimate the effect of the treatment on a) the net utility of avoiding information, and b) each of its components separately; both excluding the set of controls and fixed effects ("minimal" models) or including them ("full" models). The treatment coefficient of each regression, capturing the treatment effect on each dependent variable, is shown in Figure 9. Starting from the top of the figure, the first pair of coefficients shows that the treatment has a positive and highly significant effect on the net utility of avoiding information —plausibly, this increase is what drives the effect on choices reported in Section 5.3. How is this effect decomposed? The second pair of coefficients shows the effect of the treatment on the expected utility of finding out bad news about the state upon

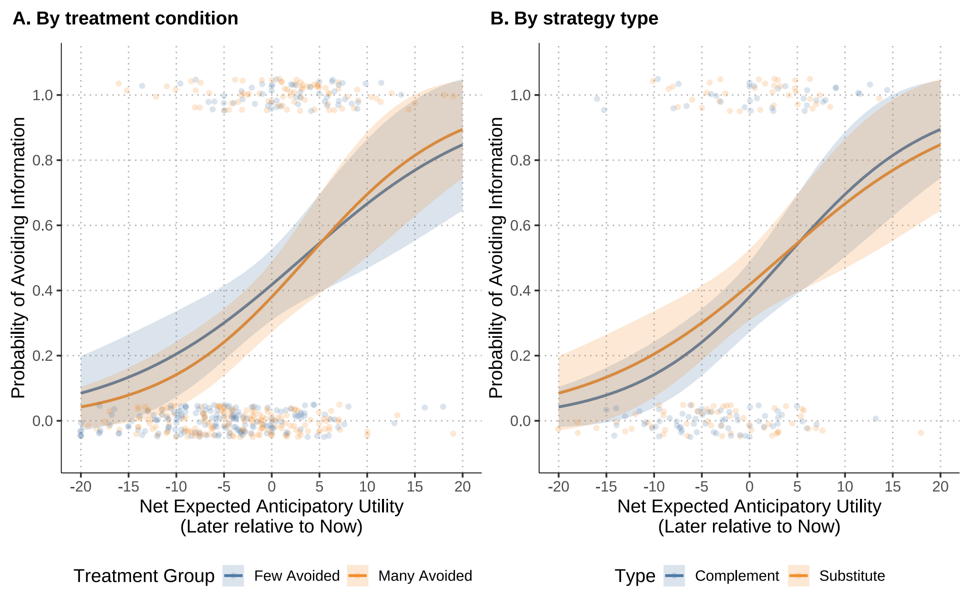


Figure 8: Association between net expected anticipatory utility and information avoidance

Notes: The figure illustrates the relationship between the net expected anticipatory utility of avoiding information and the propensity to avoid information. The net expected anticipatory utility is calculated as the difference between the utility of avoiding information and the expected utility of acquiring information, as elicited from the KP questions. Each point represents an individual participant's data. The solid line represents the predicted probability of information avoidance based on a probit regression model, estimated separately for each treatment condition (panel A) or for each strategy type (panel B), with shaded bands indicating the 95% confidence interval. The positive slope indicates that individuals with a higher net utility of avoidance are more likely to choose to avoid information.

acquiring information. The utility of discovering bad news significantly deteriorates in the *Many Avoided* condition relative to the *Few Avoided* condition. This is consistent with the model, as the payoff in the bad state (the severity of the screams) worsens as more group members avoid information. Contrastingly, as shown by the third pair of coefficients, the treatment has no effect in the expected utility of finding out good news. This again aligns with the theoretical intuition: since the payoff of the good state (silence, g) is unaffected by the number of avoiders in the group, there should be no treatment effect on the utility of discovering good news. Finally, the treatment also increases the utility of avoiding information and learning no news.

Unexpectedly, the treatment also increases the subjective belief that the state is bad. In principle, this suggests the possibility that changes in beliefs about the state in response to the treatment could be part of the mechanism, either as a competing channel or by interacting with and amplifying the groupthink mechanism, as follows: updating beliefs about the information decisions of others may deteriorate the utility of learning bad news and also induce the subject to believe that the bad state is more likely, further reducing incentives to acquire information. The plausibility of this alternative channel can be informed by the empirical literature on the relationship between prior beliefs about the state and information decisions. In the setting closest to ours, Falk and Zimmermann (2024) find that, in an individual decision-making setting, experimentally manipulating prior beliefs about the bad state does not lead to significant changes in information avoidance.

Taken together, the treatment effects on each utility component combine to increase the implied *net* anticipatory utility of avoiding information, suggesting that one of the main sources of increased incentives to avoid information in the *Many Avoided* condition (relative to the *Few Avoided* condition) arises from the deterioration of the utility of finding out bad news, a result consistent with the groupthink mechanism of Bénabou (2013) and of our model.

5.5.2 Other Mechanisms

While the evidence is consistent with the groupthink channel, the interdependence of information avoidance could, in principle, arise from other behavioral mechanisms. We discuss some of the main possibilities below.

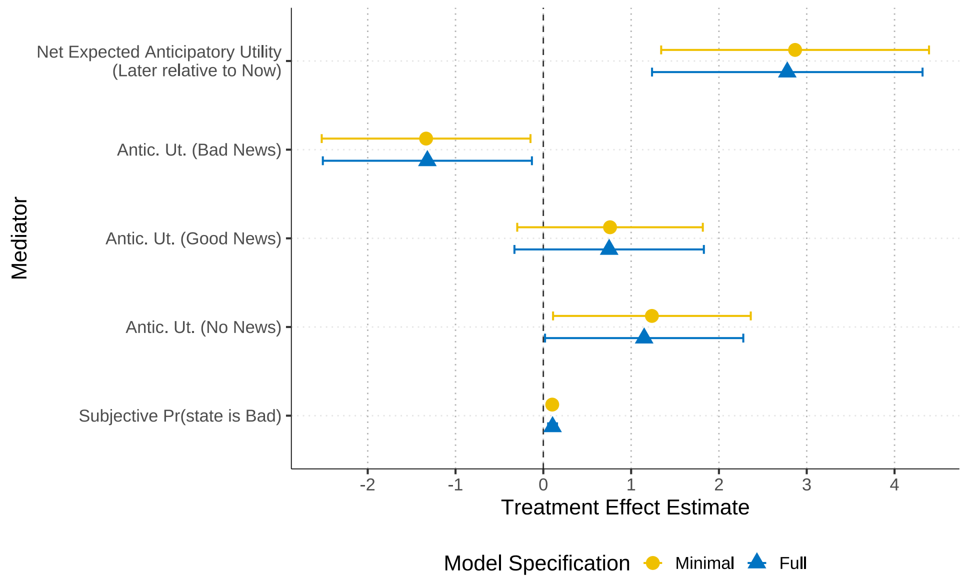


Figure 9: Mechanisms: Treatment effect on expected anticipatory utilities (KP measures)

Notes: The figure shows the treatment effect estimate on a set of variables measuring anticipatory utilities (KP measures). Each variable in this set is regressed on the treatment dummy without any controls ("Minimal" model), or with additional controls ("Full" model), including age, gender, economic and social preferences, the Information Preferences Scale, and strata and field of study fixed effects. The first dependent variable displayed on the y axis, the Net Expected Anticipatory Utility, represents the difference in expected anticipatory utilities of avoiding and acquiring information, as derived from the KP questions. This variable is decomposed by the remaining variables displayed on the y axis: the expected anticipatory utility upon finding out good news (i.e., upon acquiring information and learning that the state is Quiet), bad news (i.e., learning that the state is Screams), and no news (i.e., avoiding information); and the subjective belief that the state is bad (Screams state).

Conformity and herding. A first alternative possibility is that the positive correlation between beliefs about others' behavior and own behavior is driven by conformist preferences. We say that an individual is conformist if he derives *direct* utility from matching his actions to those of others. Alternatively, the positive correlation may be driven by information herding. Under this mechanism, subjects are unsure about which is the most convenient course of action (i.e., to avoid or acquire information) in our particular experimental setting; if they believe that the signal about others' behavior in the past is more informative about the most convenient decision than their own priors, they may rationally follow the crowd and match their actions to that of others. While each mechanism is grounded in distinct psychological motivations, they all formally predict the same empirical pattern: an individual's likelihood of avoidance increases with the perceived fraction of avoiders in their group. Given the similar empirical predictions, we label these mechanisms jointly as "conformity" for brevity. To assess them, we conduct four falsification tests. Specifically, if the observed treatment effect is purely driven by conformity, then we would expect four observable implications. First, since the treatment effectively changes the beliefs about others' information decisions, the utility of acquiring information and discovering the *good* state should differ across conditions. We focus on the event of acquiring information and discovering good news because material payoffs g are invariant in the share of avoiders, and hence, constant across conditions (unlike the payoffs in the bad state), allowing a cleaner comparison. As Figure 9 shows, we don't find support for this implication of conformity. Second, under a pure conformity channel the belief about others' information avoidance should predict own avoidance *independently* of the subjective belief about the state. In contrast, in the groupthink channel, what others do is relevant to the extent that the subject believes that the state is bad. To test these predictions, we regress $Avoid_{ig}$ on both beliefs and their interaction. We find that, contrary to the conformity channel, the interaction of both beliefs is highly significant, suggesting that the belief about others' behavior predicts own avoidance only if the subject believes that the probability of the bad state is high. Moreover, once the interaction is included, the coefficient on the belief about others' information decisions becomes insignificant. This pattern lends support to the Bénabou's (2013) mechanism rather than to conformity. Third, conformity, if effective, would be manifested in Strategic Complements only. Assuming that subjects derive utility from imitating behavior of people from *both* past and current sessions, we should observe a shift in the switching threshold of Strategic Complements caused by the treatment, given that the strategy method elicits choices conditional on the choices of group members in the current, but not past, session. As Figure 7 showed, we do not find evidence of different switching thresholds across conditions. Fourth, under a pure conformity channel the effect of the treatment should not interact with reciprocity. However, we find mild evidence of the interaction. Taken together, the four tests do not provide significant support for either the conformity or the information herding channel, while in some instances reinforcing the evidence in favor of the groupthink mechanism.

Punishment. A second potential competing mechanism is punishment. Under this mechanism, subjects in the *Many Avoided* condition expect a worse payoff in the bad state as a result of others avoiding information, and respond to this mistreatment by reciprocating

—i.e., by avoiding information, which by worsening the outcomes of other members achieves punishment. Under this mechanism, we would expect the treatment effect to be stronger for more negatively reciprocal people. In other words, the interaction between the treatment and negative reciprocity should be significant and positive. We do not find evidence of this: in regressions including interactions with prosocial preferences, the interaction between the treatment dummy and negative reciprocity is insignificant.

Social Learning. A third competing mechanism is social learning. Under this mechanism, subjects do not know the parameters of their preference over screams, and use the signal about behavior in the past to update their beliefs about their own preference parameters. As a result, this may generate differences in information avoidance across treatment conditions. Although we do not have a direct test for this channel, we expect this mechanism to be negligible.

Non-linear preferences over volume. Finally a fourth competing mechanism relates to differences in the perceived benefit of a reduction of the 2 volume points across conditions. If subjects perceive that a 2 point reduction is less valuable in the *Many Avoided* condition relative to the *Few Avoided* condition, they would have higher net incentives to avoid information. While this mechanism is possible in principle, casual experience suggests that an increase of 2 volume points in a computer (out of 100 points) generates only a subtly noticeable difference in perceived loudness. This suggests that the effect through this mechanism may be negligible, unless subjects exaggerate their expectations of changes in loudness both substantially and asymmetrically between conditions.

6 Group Composition and Equilibria

While the main treatment effect suggests that exogenously increasing beliefs about the prevalence of avoidance increases the average likelihood of avoidance in a group, we have also documented heterogeneity in individuals' best response schedules to others' avoidance. This heterogeneity suggests that depending on the group's composition of strategy types, the *equilibrium* prevalence of information acquisition may change. An "information equilibrium" is defined as an equilibrium fraction of information avoiders in a group. In other words, a share of avoiders π^* is an information equilibrium if a proportion π^* of group members best-responds to a share π^* by avoiding information and $1 - \pi^*$ best-responds by acquiring information. This section explores the relationship between group composition and the resulting set of information equilibria.

To derive the set of information equilibria of a group g, we construct the group-level, or "aggregate", best response function by aggregating the individual best-response functions of its members, as follows. The group is characterized by N_g individual best-response schedules, each defined as a function $f_{gi} : [0, 1] \to \{0, 1\}$ that maps each possible share of avoiders to a decision to avoid (1) or acquire (0) information. To derive the aggregate best response function F_g, we simply take the average of f_{gi} at each π, over group members. For example, consider a simple group composed by one Always Getter (with $f_{g1}(\pi) = 0$ for all π) and one Always Avoider (with $f_{g2}(\pi) = 1$ for all π). Then the aggregate best response function is $F_g(\pi) = 0.5$ for all π. In other words, for any proportion of avoiders π, half of the group members would best-respond by avoiding information.

Because of implementation constraints, it is not possible to elicit f_i at *every* possible value of π: f_{gi} can be practically elicited only for a coarse set of values of π. This necessarily leads to a coarse representation of F_g. To estimate the full shape of the group-level best response function F_g, we input the missing values by linear interpolation. This method has the advantages of simplicity and of providing good local approximations in the neighborhood of the observed values of F under a continuity assumption. It also ensures that F varies smoothly with π, avoiding estimating the location of thresholds leading to discountinuous jumps.

Once the full shape of F_g is estimated, it is straighforward to identify the information equilibria of the group: it is the set of fixed points of F_g, namely, the set of points π^* such that $F_g(\pi^*) = \pi^*$, or where F_g crosses the identity line.

To gain intuition, consider "homogeneous" groups, namely, groups composed by a single strategy type. Figure 10 shows the shape of F_g for such homogeneous groups g by conditioning our empirical sample on each strategy type. For each type, blue dots represent the observed values of F_g and blue lines the segments estimated by linear interpolation. The set of equilibria, indicated by the orange diamonds, is found at the intersection(s) of F_g and the identity line, represented by the dotted diagonal line. Homogeneous groups composed by unconditional types (Always Getters or Always Avoiders) exhibit constant aggregate best reaction functions, since the decisions of its members do not react to others' information decisions. Those groups have a unique information equilibrium at the extreme of the support, where either everybody acquires (Always Getters) or everybody avoids (Always Avoiders) information. The bottom panels show homogeneous groups composed by conditional types. The group of Strategic Substitutes (Complements) exhibits a negatively (positively) sloped F_g. The group of Strategic Substitutes exhibits a unique, interior equilibrium. Notably, the group of Strategic Complements exhibits multiple equilibria, including a fully informed and a fully uninformed equilibrium, as well as an interior equilbrium, in line with the theoretical result in Bénabou (2013).

Next, we consider how all strategy types are combined in the whole empirical sample. Figure 11 shows the estimated F_g for a group composed by our whole sample. Perhaps unexpectedly, the aggregate F_g, is somewhat U-shaped, although almost constant. The flatness of F_g arises from two sources. First, around half of the individuals are unconditional types (either Always Getters or Always Avoiders), and hence contribute flat individual schedules f_{gi}'s to the aggregation into F_g. Second, while 40% of subjects are conditional types who could in principle break the flatness of F_g, these types are split equally into Complements and Substitutes, so that, roughly speaking, they average out each other. As a result, the group exhibits an almost horizontal aggregate F_g, with a unique interior equilibrium at $\pi^* = 0.30$.

Nonetheless, the diversity of shapes in aggregate reaction functions F_g shown in Figure 10 and the particular mix of types in our empirical sample highlights how we may encounter very different sets of information equilibria depending on the specific composition of the group. This motivates a more detailed study of how variations in group composition impacts the set of informational equilibria. This could be useful, for example, if a principal of an organization wants to assemble a team of N_g members and is interested in some feature of the distribution of the set of informational equilibria, such as the median, the worst, or the best informational equilibrium in the set. She may also want to bound the probability of $\pi^* > \bar{\pi}$ for some critical level of interest $\bar{\pi}$ by optimally choosing her team composition. To study this, we employ simulations using bootstrap sampling to generate randomly drawn groups of a given composition to determine the distribution of equilibria corresponding to that particular group composition, and then track how the distribution of equilibria changes with changes in the group composition, and how fast.

The algorithm is described as follows. First, fix a group size N_g. Fix a sequence of group compositions $\tau \equiv \{\tau_t\}_{t=1,2,\ldots,T}$, where each group composition τ_t is a vector $\tau_t \equiv (p_{AG}, p_{AA}, p_{SC}, p_{SS})_t$ whose elements are convex probabilities (respectively, for Always

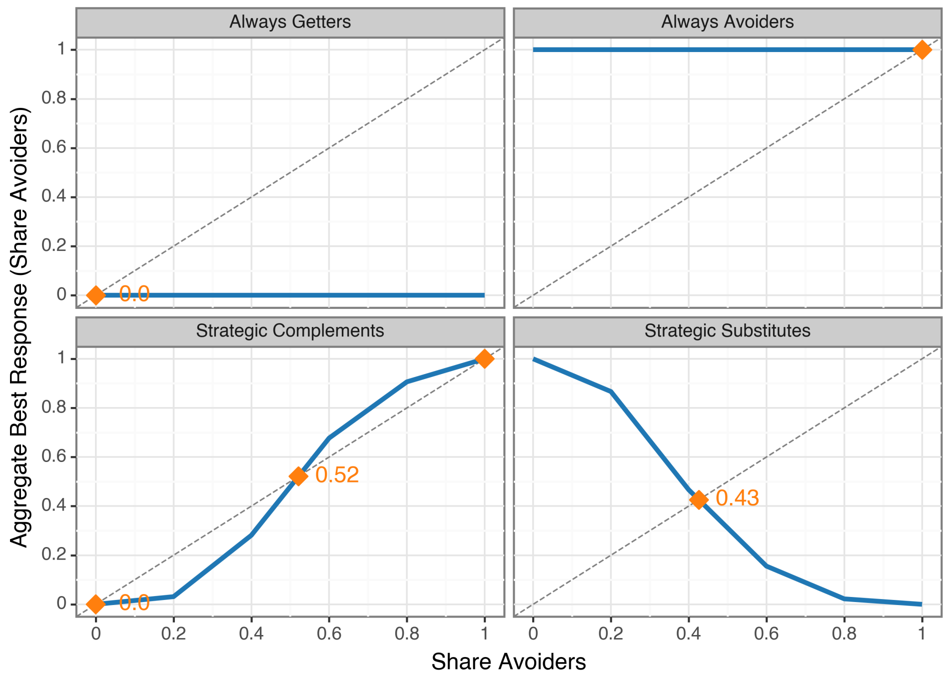


Figure 10: Aggregate Best Response Functions and Information Equilibria (Homogeneous Groups)

Notes: Aggregate (group-level) best response schedules of "homogeneous" groups composed by a single strategy type. Each group is formed by subsetting the experimental sample on the type indicated by the caption. Blue dots represent values of the schedule at elicited values of π. Blue lines show the linear interpolation between blue points. The dotted, diagonal line shows the identity line. Orange diamonds represent the fixed points of the schedule (i.e., the information equilibria).

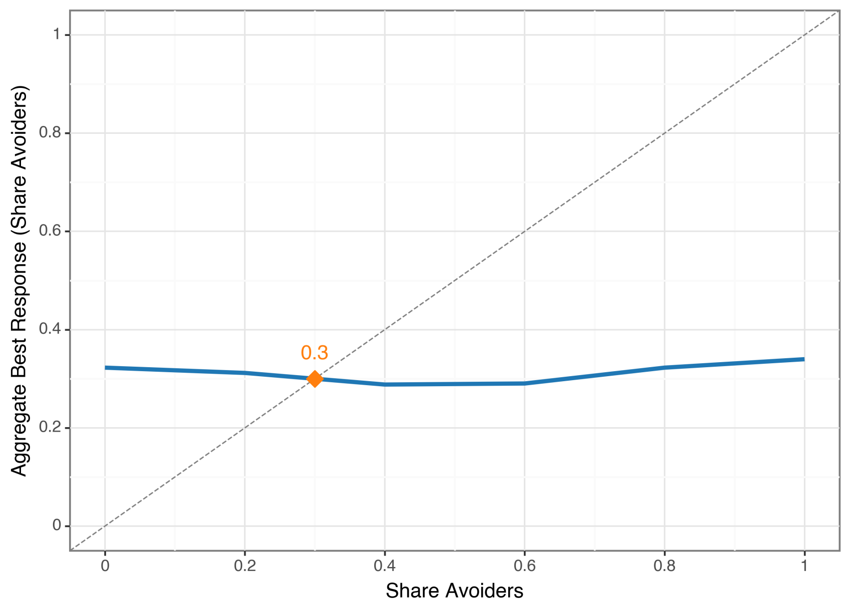


Figure 11: Aggregate Best Response Function and Information Equilibria (Full Empirical Sample)

Notes: Aggregate (group-level) best response schedule of a group composed by whole empirical sample. Blue dots represent values of the schedule at elicited values of π. Blue lines show the linear interpolation between blue points. The dotted, diagonal line shows the identity line. Orange diamond represents the fixed point of the schedule (i.e., the information equilibrium).

Getters, Always Avoiders, Complements, and Substitutes) that add up to 1. Start with $t = 1$, and the first composition τ_1. For each bootstrap iteration $b = 1, ..., B$, draw a random group of size N_g through bootstrap sampling with replacement, where the sampling probabilities of each strategy type given by τ_1. This generates B simulated groups of size N_g with composition τ_1. Among those B groups, for each group b, construct the aggregate best response F_b (including the linear interpolation) and find the set of fixed points Π_b, classifying their stability. Then, obtain composition-level statistics (computed across bootstrapped groups b): mean and 95% confidence interval of F_b at each π, and the frequency of each equilibrium share π^*. After this, let $t := t + 1$ and iterate for group composition τ_t until $t = T$.

For the simulations, we iterate over a grid of all possible compositions subject to the constraint that the proportion of types take on discrete values from 0 to 1 in steps of 0.1. Given the additional constraint that the probabilities add up to 1, the grid includes 286 different group compositions. For each composition, we bootstrap $B = 100$ groups, each with $N_g = 13$ members.[31]. In principle, it is also possible to study a large N_g, which would result in more precise estimates, but on the other hand, for N_g very large it would become unlikely for a principal to be able to control the group composition. Hence, we focus on finite group size properties. In total, the simulation involves 367,600 individuals across 28,600 groups, across 286 group compositions.

Among the large number of group compositions, consider first the sequence of compositions that starts with groups composed entirely by Complements (which exhibit the highest multiplicity of equilibria and the highest uncertainty) and that gradually increase the proportion of either Always Getters or Always Avoiders. Specifically, we consider two sequences: one starts from 100% Complements and increases the proportion of Always Getters by 10% in each step t, and the other starts from 100% complements and increases the proportion of Always Avoiders by 10% in each step t. Therefore, we study how the introduction of Getters or Avoiders can move a group with multiple equilibria towards a unique equilibrium.

Figure 12 shows the evolution of the average shape of aggregate best reaction functions as the composition changes from fully conditional types to fully unconditional types. The

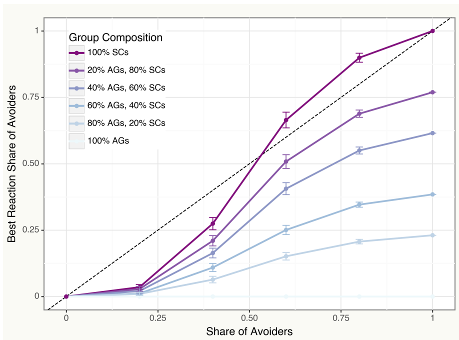
[31]If N_g is a "round" number, such as 10, it is possible that, for some draws, continuous portions of F_b lie exactly on top of the identity line, introducing an infinite number of equilibrium points. This introduces technical details about how to count and adjust for these when calculating the relative frequency of equilibria, as well as apparent non-monotonicities in features of the distribution of equilibria as the group composition τ_t varies. To avoid these issues, we focus the discussion on cases constructed to prevent the emergence of infinite number of equilibria.

overall pattern is that the inclusion of unconditional types flattens the aggregate best reaction function and pushes it towards the extremes, namely either 0 (full Always Getters) or 1 (full Always Avoiders). Therefore, as more unconditional types are represented in the group, interior equilibria should become less likely.

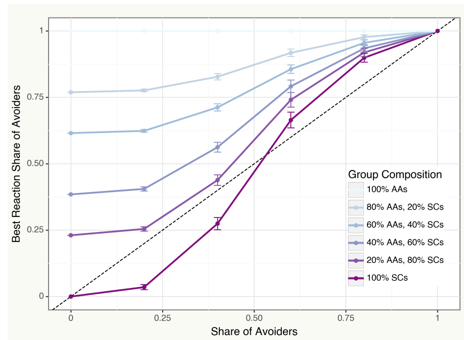
In fact, Figure 13 shows the distribution of fixed points (i.e., the information equilibria) across the compositions in the sequence. Groups of pure Strategic Complements have a trimodal distribution, since Complements typically exhibit both extreme and an interior equilibrium. The two corner equilibria tend to be stable, whereas interior equilibria tend to be unstable. On the other hand, groups of pure Substitutes exhibit only interior equilibria, which are typically unstable. For both Complements and Substitutes, as the group changes its composition towards a higher representation of Always Getters (Avoiders), the distribution of equilibria shifts to the left (right).

Since, in practice, unstable equilibria are unlikely to persist and to be observed, in what follows we restrict attention to the set of stable equilibria. So far, to visualize variations in composition, we considered sequences that restricted the probabilities of two types to be zero. Now, since Substitutes are highly likely to produce unstable equilibria, we omit them in subsequent analyses, so that compositions are redefined as 3-dimensional vectors of probabilities. This, in turn, allows representing the *entire* space of compositions in a unit probability triangle. Figure 14 shows such triangle. The proportion of Always Getters (Always Avoiders) is represented on the $x(y)$-axis; the proportion of Complements is implied by the complementary probability and visible as the horizonal (or vertical) distance from the cell to the hypothenuse. Each cell is associated to a composition, and all compositions are identified by a cell in this triangle. Each cell, in turn, has associated a distribution of fixed points (for some cells, the distributions have been depicted by the histograms above). A policy-maker may care about different *features* of such distributions, such as the mean, the median, or the variance. Panel (a) presents, for each cell, the average among the set of all stable equilibria. A group of pure Complements (the cell at $(0, 0)$) has an average close to 0.5, since the distribution of fixed points is near symmetric. As we add Always Getters (moving horizontally from (0,0) along the x axis) the average equilibrium share of avoiders decreases smoothly, down until zero. Conversely, as we add Always Avoiders (moving vertically from (0,0) along the y axis) the average equilibrium share of avoiders increases smoothly, up until unity. In general, across the triangle, changes in the average fixed point appear to be smooth.

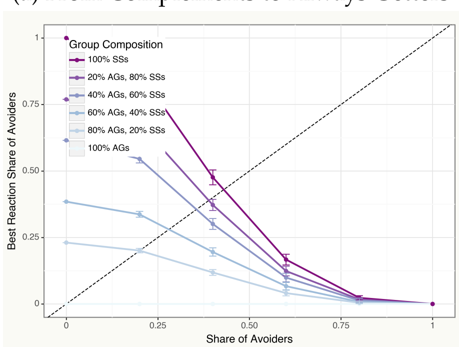
Panel (b) shows the maximum in the set of equilibria, for each cell. Intuitively, if a



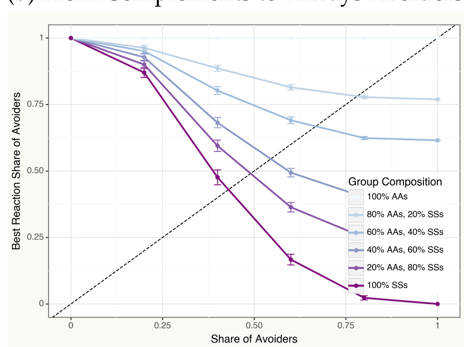
(a) From Complements to Always Getters



(b) From Complements to Always Avoiders



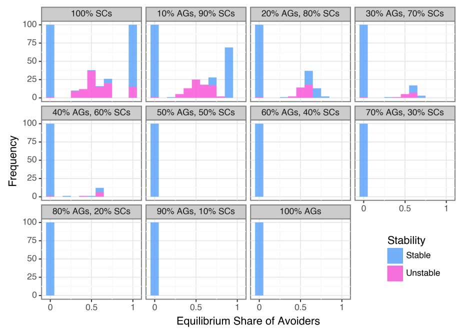
(c) From Substitutes to Always Getters



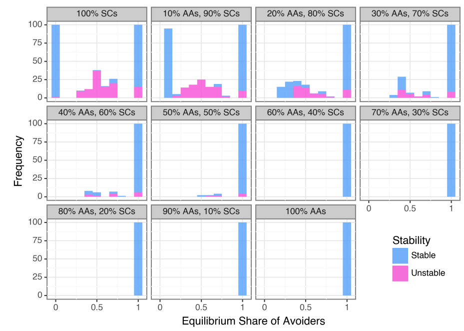
(d) From Substitutes to Always Avoiders

Figure 12: Effect of gradual changes in group composition on shape of aggregate best response function

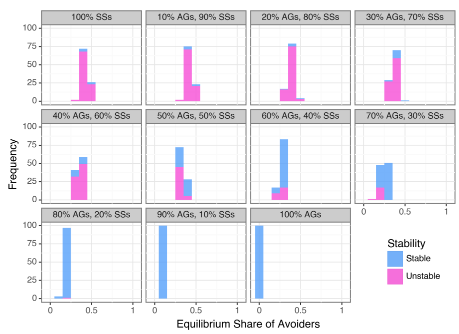
Notes: The figure shows the effect of gradual changes in group composition on the shape of the aggregate best response function. Each line depicts the average (across bootstrapped groups) aggregate best response schedule for a given group composition. In each panel, the darkest purple line represents groups composed purely by a conditional type (Strategic Complements for the top row and Strategic Substitutes for the bottom row). Lines with lighter colors represent gradual composition changes towards a higher representation of unconditional types (Always Getters for the first column, Always Avoiders for the second column). Error bars represent 95% confidence intervals.



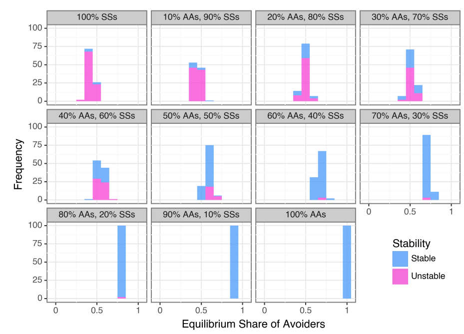
(a) From Complements to Always Getters



(b) From Complements to Always Avoiders



(c) From Substitutes to Always Getters



(d) From Substitutes to Always Avoiders

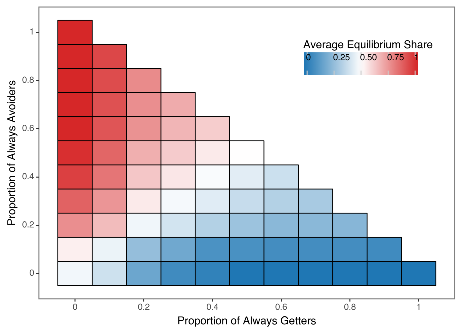
Figure 13: Effect of gradual changes in group composition on distribution of information equilibria

Notes: The figure shows the distribution of fixed points (i.e., shares of avoiders that are information equilibria in the corresponding bootstrapped group) by composition. Each facet corresponds to a composition. Bars represent absolute frequencies (counts) of each fixed point value on the x-axis. The color of the bars represent the stability of the fixed point: stable (unstable) equilibria are represented by blue (pink) bars.

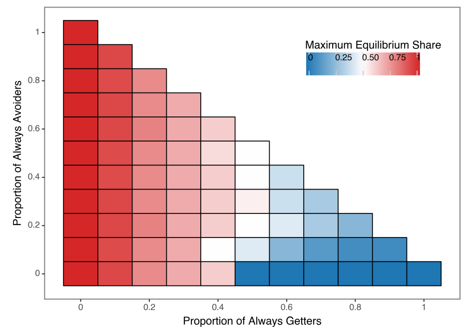
policy-maker considers avoidance of information to be an undesirable behavior, panel (b) shows what is the "worst" equilibrium in the distribution, for each cell. The worst equilibrium of group of pure Complements is full avoidance, and adding Always Avoiders doesn't change the worst equilibrium. However, adding Always Getters improves the worse equilibrium: the maximum in the distribution decreases monotonically. An interesting pattern that emerges is a strong discontinuity around $p_{AG} = 0.5$. More generally, there are regions of discontinuity across the space of compositions, where a a fixed amount of change in composition drastically moves the feature of interest. The discontinuities cluster around $p_{AG} = 0.5$: when the proportion of Always Getters is sufficiently high, the aggregate schedule shifts sufficiently downwards, such that (stable) interior equilibria are eliminated. The discontinuities are weaker the higher the proportion of Always Avoiders, as the presence of this type helps sustain interior equilibria. Panel (c) shows the minimum among the equilibria (the "best" equilibrium in the distribution), for each cell, with patterns analogous to those for the maximum. Now, the relevant probability is p_{AA}. Finally, a policy-maker may be interested in the variance of the set of possible equilibria (for example, through risk-aversion). Panel (d) depicts the variance. As expected, the variance is maximal in groups with of pure Complements, which feature multiple equilibria. As unconditional types are added, the schedule becomes flatter, and the distribution of equilibria starts to degenerate.

To visualize nonlinearities more clearly, we focus again on "restricted" group compositions tracing changes from an interdependent type (either Complements or Substitutes) towards an independent type (either Always Getters or Always Avoiders). For each of these restricted sequences of compositions, Figure 15 shows how a number of features of the distribution (the mean, median, max, min, and other quantiles) respond to changes in the composition. The overall pattern is that features of the distribution of equilibrium points appear to respond monotonically, but non-linearly, to changes in the composition. Moreover, for some features such as the maximum and the minimum, there are regions in the composition sequence where the feature responds sharply, suggesting that the gains from a fixed amount of change in composition can vary across the composition space.

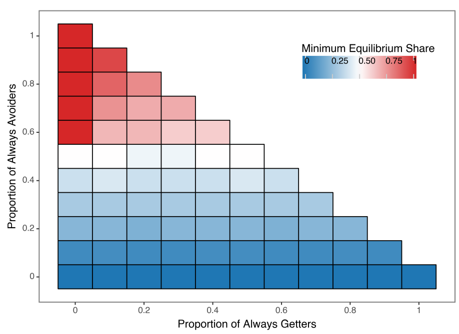
The simulations, depicted in the figure, also inform probabilistic predictions about the likelihood of observing a given equilibrium level of avoidance: for example, starting from a group of 100% Complements, it suffices to move the composition to 30% Always Getters and 70% Complements to make full information acquisition equilibrium occur with 75%



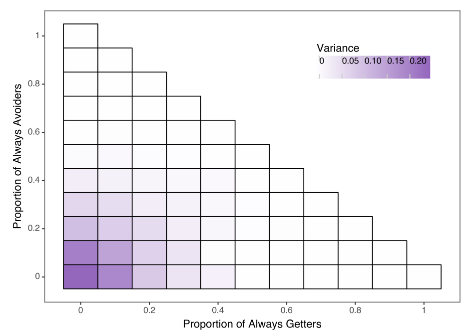
(a) Average of equilibria



(b) Maximum among equilibria



(c) Minimum among equilibria



(d) Variance of equilibria

Figure 14: Group compositions and features of the distribution of information equilibria

Notes: The figure shows the distribution of fixed points (i.e., shares of avoiders that are information equilibria in the corresponding bootstrapped group) by composition. Each facet corresponds to a composition. Bars represent absolute frequencies (counts) of each fixed point value on the x-axis. The color of the bars represent the stability of the fixed point: stable (unstable) equilibria are represented by blue (pink) bars.

probability (as shown by the 75% quantile). Moving the composition further to 50% Always Getters and 50% Complements ensures the probability of a full information acquisition equilbrium is 100%.

7 Conclusion

We live in an era of abundance of information in a variety of domains —politics, health, financial, environmental. Despite the wide availability of virtually free information, sometimes individuals prefer to avoid it, choosing, in effect, to remain in a state of willful ignorance. In this paper, we study whether this behavior is interdependent among individuals whose information decisions generate payoff externalities on other members. Our experimental evidence suggests that, on average, willful ignorance can be contagious: individuals are more likely to avoid readily available information about the state when they expect others to do so, and hence, when they expect the outcome in the bad state to be more severe (and news revealing that the state is bad to be more aversive). Our findings suggest that the interdependence is mediated by an increase in expected anticipatory disutility of discovering bad news as a result of others' information avoidance, and it is moderated by social preferences, particularly, by reciprocity. However, we document substantial heterogeneity in individual strategy types: while most individuals' optimal information decisions are independent from others' decisions, about 40% the subjects exhibit interdependence —they condition their information decisions on those of others. This heterogeneity of reaction functions suggests that the degree of aggregate contagiousness of willful ignorance may depend crucially on both the composition of the group in terms of reaction functions and the distribution of subjective expectations about others' behavior. Simulations show that, in effect, changes in group composition generate a wide variety of information equilbria, ranging from full information acquisition (and maximum mitigation of the bad state outcome) to full willful ignorance (and maximum severity).

Our findings add nuance to the recommendations from the literature on information preferences, which typically focus on individual decision settings. Our results suggest that incentivizing a person to acquire information (which may be beneficial for the individual in isolation) may generate positive or negative informational externalities to other members, in the form of stronger or weaker incentives to acquire information. Therefore, policy-makers should take into account these "cognitive externalities" and general equilibrium effects when

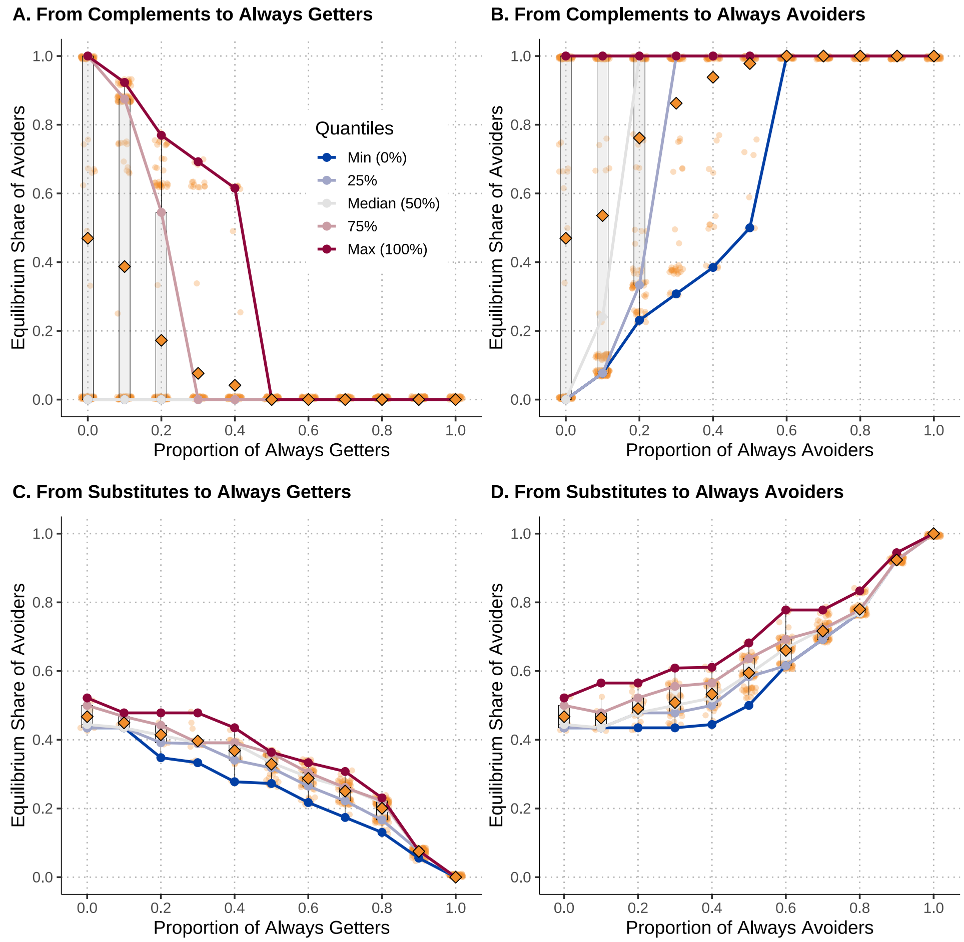


Figure 15: Effect of Group Composition on Distribution of Information Equilibria

Notes: This figure illustrates the effect of gradual changes in group composition on the distribution of information equilibria. The x-axis represents the share of a specific strategy type, while the y-axis indicates values for the equilibrium share of avoiders. Each panel depicts the equilibria from bootstrap samples: individual equilibria (orange dots), their mean (orange diamonds), and key distributional characteristics shown via boxplots (gray boxes; interquartile range) and quantiles (min/max in blue/red dots, median in gray dots). **Panel A** shows the results of gradually replacing Strategic Complements with Always Getters, and **Panel B** with Always Avoiders. **Panel C** replaces Strategic Substitutes with Always Getters, and **Panel D** with Always Avoiders.

designing policies that aim at increasing the aggregate take-up of information in groups when individuals' outcomes are interlinked.

Our results suggest some implications for organization design. Given that, usually, in organizations, information has instrumental value (i.e., subsequent decisions can be adjusted and improved if information about the state is acquired), a principal may be interested in incentivizing acquisition of information in her organization in order to minimize the emergence of a groupthink equilibrium where information avoidance is prevalent. Taking the group as given, a potential way to encourage information take-up is by correcting potential misperceptions about how others deal with information [EXPAND]. Another way is to design the group composition in terms of strategy types when assembling teams. Our simulations show that the group composition has a strong impact in the distribution of equilibrium shares of informed members, and suggest the equilibrium responses are non-linear in the group composition, in a way that there are regions in which the same amount of composition change delivers maximum results.

Our investigation has some limitations and motivates avenues for future research. First, because our experimental setup is based on the setup of Bénabou (2013) where there is a theoretical link between agents' decisions and the severity of the outcome, our design cannot disentangle whether (and how much of) the contagiousness arises from variations in the severity of the outcome ("severity effect") or from variations in the proportion of others who engage choose to remain ignorant ("herding effect"). This limitation applies both to the average treatment effect and the individual reaction functions. We view our study as a first step in establishing the empirical existence of interdependence in information decisions. Further research is needed to disentangle the severity effect from the herding effect. Second, we consider a static decision-making setting, where all individuals choose their information decisions simultaneously. In many applications, such decisions occur sequentially. Studying sequential decisions, however, introduces experimental design challenges, such as reputation concerns and (possibly non-Bayesian) inference. Future studies could explore willful ignorance in groups in dynamic settings.

Our study highlights that understanding the ways in which individuals react to others' information decisions is crucial for designing environments that promote the acquisition of information, which, usually, improves subsequent decision-making in organizations.

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TLDR: The updated and correct Figure 4 from "Optimal Expectations and Limited Medical Testing: Evidence from Huntington Disease" shows that for low symptom levels individuals are correct about their risk level, whereas those with more advanced symptoms are overly optimistic.

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Appendices

A Proofs

Proof of Lemma 1

Consider v_i everywhere convex. Agent i prefers to avoid information if $\varphi_i > 0$, i.e.

$$v_i\left(pb + \left(1 - p\right) g\right) > pv_i\left(b + \Delta\right) + \left(1 - p\right) v_i\left(g\right)$$

Since $v_i\left(.\right)$ is strictly increasing, we have $v_i\left(b + \Delta\right) > v_i\left(b\right)$, hence we must have

$$v_i\left(pb + \left(1 - p\right) g\right) > pv_i\left(b\right) + \left(1 - p\right) v\left(g\right)$$

but this contradicts convexity. Therefore, the agent never avoids information, i.e., the agent is an Always Getter.

Proof of Lemma 2

Suppose v_i everywhere concave. First note that the net incentive to avoid information increases in the share of other group members who avoid information:

$$\begin{aligned}\frac{\partial \varphi_i}{\partial \lambda_{-i}} &= pv_i'\left(pb + \left(1 - p\right) g\right) \frac{\partial b}{\partial \lambda_{-i}} - pv_i'\left(b + \Delta\right) \frac{\partial b}{\partial \lambda_{-i}} \\ &= \left[v_i'\left(pb + \left(1 - p\right) g\right) - v_i'\left(b + \Delta\right)\right] p \frac{\partial b}{\partial \lambda_{-i}} \\ &> 0\end{aligned}$$

by concavity of v_i and assumption 1. It follows that:

- If $\varphi_i\left(\lambda_{-i} = 0\right) > 0$ then by monotonicity $\varphi_i\left(\lambda_{-i}\right) > 0$ for any λ_{-i}, that is, the agent is an Always Avoider.
- If $\varphi_i\left(1\right) < 0$ then by monotonicity $\varphi_i\left(\lambda_{-i}\right) < 0$ for any λ_{-i}, that is, the agent is an Always Getter.
- If $\varphi_i\left(0\right) < 0$ and $\varphi_i\left(1\right) > 0$, then by continuity there exists an interior λ_{-i}^* such that the agent acquires information whenever $\lambda_{-i} < \lambda_{-i}^*$ and avoids it when $\lambda_{-i} > \lambda_{-i}^*$ (strategic

complementarity).

Proof of Lemma 3

We assume throughout that $b(0) + \Delta < \widehat{U}_i$. Suppose that if everyone else avoids information, then at date 1 the expected utility under ignorance is so low that it falls below the reference level: $pb(1) + (1-p)g < \widehat{U}_i$. Then v_i is convex at $pb(1) + (1-p)g$. By convexity,

$$v_i\left(pb(1) + (1-p)g\right) < pv_i(b(1)) + (1-p)v_i(g)$$

so even without instrumental value of information Δ, avoiding information is dominated by acquiring it. Since v_i increasing, adding the instrumental value makes avoiding information even less desirable:

$$v_i\left(pb(1) + (1-p)g\right) < pv_i(b(1)) + (1-p)v_i(g) < pv_i(b(1) + \Delta) + (1-p)v_i(g)$$

Therefore, $\varphi^i(1) < 0$. When all other agents acquire information, agent i prefers to avoid. Suppose that if everyone else acquires information, then at date 1 the expected utility under ignorance goes above the reference level: $pb(0) + (1-p)g > \widehat{U}_i$. Then v_i is concave at $pb(0) + (1-p)g$. By concavity,

$$v_i\left(pb(0) + (1-p)g\right) > pv_i(b(0)) + (1-p)v_i(g)$$

For avoidance to be optimal, we need

$$v_i\left(pb(0) + (1-p)g\right) > pv_i(b(0) + \Delta) + (1-p)v_i(g)$$

Define the difference in expected utilities $d \equiv v_i(g) - v_i\left(pb(0) + (1-p)g\right)$. Then, the avoidance condition can be re-expressed as:

$$\begin{aligned} v_i(g) - d &> pv_i(b(0) + \Delta) + (1-p)v_i(g) \\ \text{i.e. } p\left[v_i(g) - v_i(b(0) + \Delta)\right] &> d \end{aligned}$$

When $d \to 0$, this always holds. Therefore, by continuity, there exists a value $d^* > 0$ such that for $d < d^*$ the agent avoids information. If $d < d^*$, the agent avoids information when all other agents in the group acquire it ($\varphi_i(\lambda_{-i} = 0) > 0$) and acquires information when all others avoid it ($\varphi_i(\lambda_{-i} = 1) < 0$). By continuity of $\varphi_i(\lambda_{-i})$, there exists an interior threshold λ^{**} such that agent i strictly prefers to avoid information if $\lambda_{-i} < \lambda_i^{**}$ and strictly prefers to acquire information if $\lambda_{-i} > \lambda_i^{**}$ (strategic substitutability).

Proof of Lemma 4

An altruistic agent with altruism parameter $\alpha_i > 0$ avoids information if

$$\varphi_i \equiv v_i\left((1+\alpha_i)\left(pb + (1-p)\,g\right)\right) - pv_i\left((1+\alpha_i)(b+\Delta)\right) - (1-p)\,v_i\left((1+\alpha_i)\,g\right) > 0$$

Increasing altruism increases her net incentive to avoid information iff $\partial\varphi_i/\partial\alpha_i > 0$:

$$\left(pb + (1-p)\,g\right)v_i'\left((1+\alpha_i)\left(pb+(1-p)\,g\right)\right) > p\,(b+\Delta)\,v_i'\left((1+\alpha_i)(b+\Delta)\right) + (1-p)\,gv_i'\left((1+\alpha_i)\,g\right)$$

which can be rearranged as

$$\begin{aligned} & pb\left[v_i'\left((1+\alpha_i)\left(pb+(1-p)\,g\right)\right) - v_i'\left((1+\alpha_i)(b+\Delta)\right)\right] \\ & + (1-p)\,g\left[v_i'\left((1+\alpha_i)\left(pb+(1-p)\,g\right)\right) - v_i'\left((1+\alpha_i)\,g\right)\right] \\ & > p\Delta v_i'\left((1+\alpha_i)(b+\Delta)\right) \end{aligned}$$

The RHS captures the fact that more altruistic agents have stronger instrumental motives to acquire information, since they internalize more the positive externalities of information acquisition on other members. Since $v' > 0$, the RHS is strictly positive. Thus a necessary condition for $\partial\varphi_i/\partial\alpha_i > 0$ is LHS > 0.

For v_i everywhere concave, the expression in the second square bracket is strictly positive, since $g > b$. The expression in the first square bracket is strictly negative, since $\alpha_i > 0$ and given the assumption on Δ. Therefore:

- If $b > 0$ then the sign of the LHS is ambiguous.
  - If $p = 0$, then $\partial\varphi_i/\partial\alpha_i > 0$, so more altruistic types have stronger incentives to avoid.
  - If $p = 1$, then $\partial\varphi_i/\partial\alpha_i < 0$, so more altruistic types have weaker incentives to avoid.

- If $b < 0$, then the LHS> 0.
  - If $p = 0$, then $\partial\varphi_i/\partial\alpha_i > 0$
  - If $p = 1$, then the sign of $\partial\varphi_i/\partial\alpha_i$ is ambiguous.

[Proof to be completed]

B Are Screams a Bad? Preferences for Screams and Volume

We identify Volume Haters (Lovers) as those subjects who report a higher (lower) utility of hearing screams at volume 50 than at volume 100. We identify Scream Haters (Lovers) as those subjects who report a higher (lower) utility of finding out that the state is Quiet than that it is Screams. In addition, for each individual we obtain a measure of the strength of the hate for volume by taking the difference between the utility of hearing screams at volume 50 and the utility of hearing screams at volume 100. Similarly, we obtain a measure of the strength of the hate for screams by taking the difference between the utility of finding out the Quiet state and the utility of finding out the Screams state.

Figure B.1 presents the joint distribution of subjects' reported utilities of hearing screams at different volumes, identifying each type of volume preference. Figure B.2 presents the joint distribution of subjects' reported utilities of learning each state, identifying each type of scream preference. The distribution of the *strength* of hate for screams (volume) is shown in Figure B.3 (Figure B.4).

The joint distribution of preference types over screams and volume is shown in Table B.1.

	Volume Preference Type				
Scream Preference Type	**Volume Lover**	**Volume Indifferent**	**Volume Hater**		**Total**
Scream Lover	23	3	34	4	64
Scream Indifferent	1	5	19	1	26
Scream Hater	16	5	330	24	375
Total	40	13	383	29	465
Scream Lover	4.9%	0.6%	7.3%	0.9%	13.8%
Scream Indifferent	0.2%	1.1%	4.1%	0.2%	5.6%
Scream Hater	3.4%	1.1%	71.0%	5.2%	80.6%
Total	8.6%	2.8%	82.4%	6.2%	100.0%

Table B.1: Distribution of Scream- and Volume-Preference Types

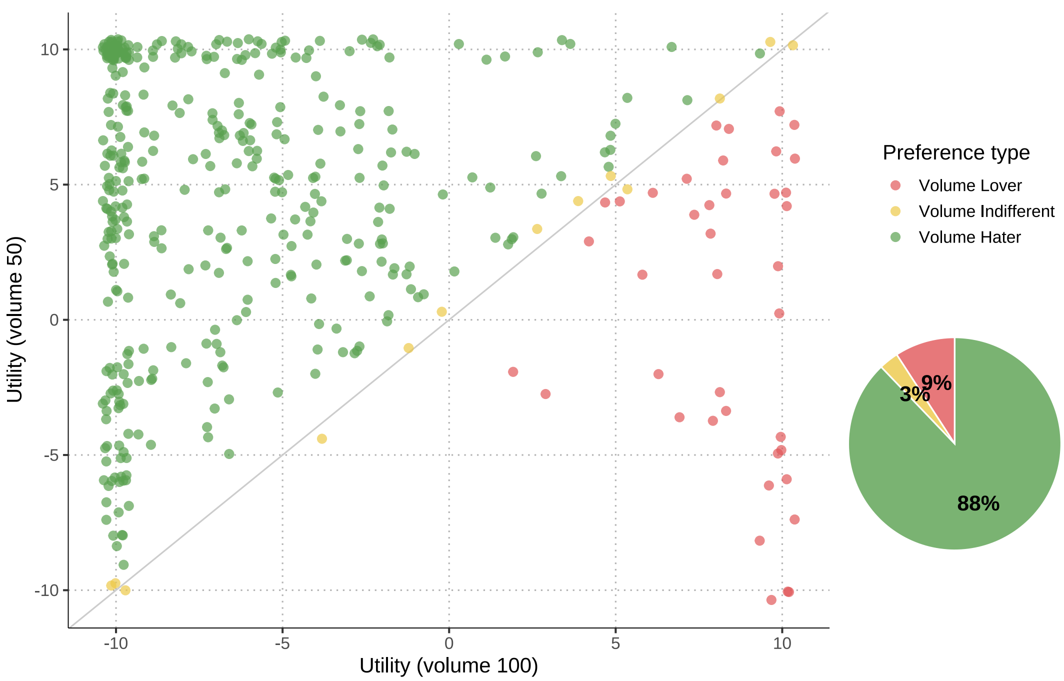

Figure B.1: Preferences over scream volume: reported utilities if screams were played at different volumes

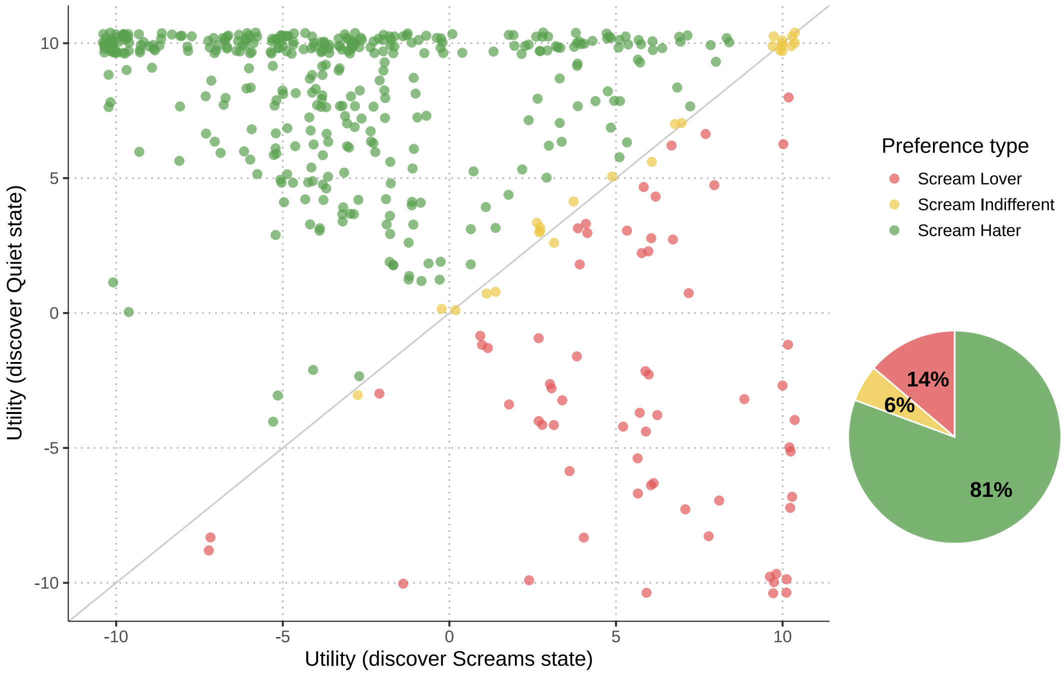

Figure B.2: Preferences over screams: Reported utilities upon learning each state

B.1 Robustness to Exclusion of Scream- and Volume-Lovers

The main regression results are robust to the exclusion of Scream- and Volume-Lovers. Figure B.5 shows the treatment effect on information avoidance by type of preference for scream (panel a) and for volume (panel b). The treatment effect is positive for scream- and volume-haters, but the direction of the effect is, if anything, the opposite for scream- and volume-lovers. This suggests that, by not excluding scream- and volume-lovers from our main regressions, the treatment effect is rather attenuated.

Table B.2 shows our main regression results if we exclude Scream-Lovers. Table B.3 shows our main regression results if we exclude Volume-Lovers. In both subsamples, the magnitude of the estimated treatment effect increases, compared to the full sample (reported in Table 2), for all the reported specifications. The treatment coefficient becomes significant at 1% for all specifications.

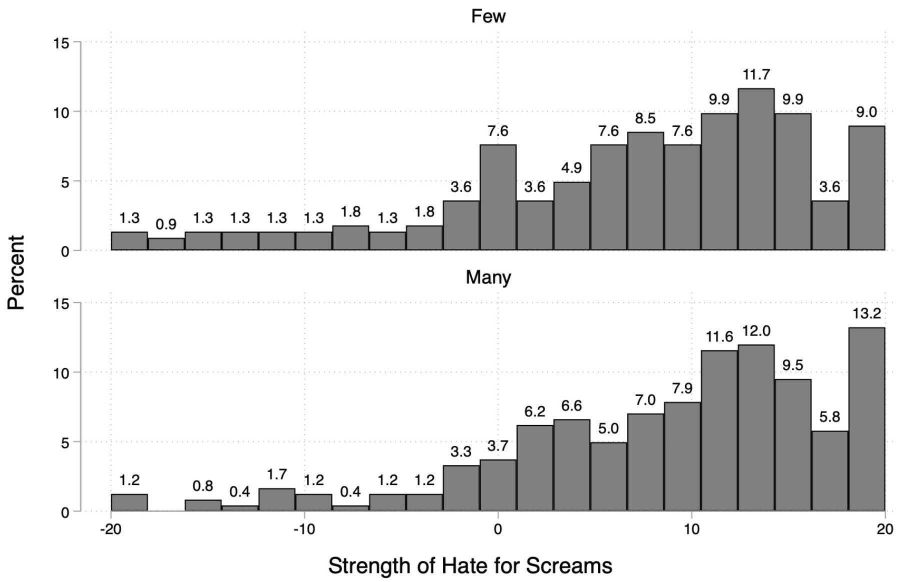

Figure B.3: Strength of Hate for Screams

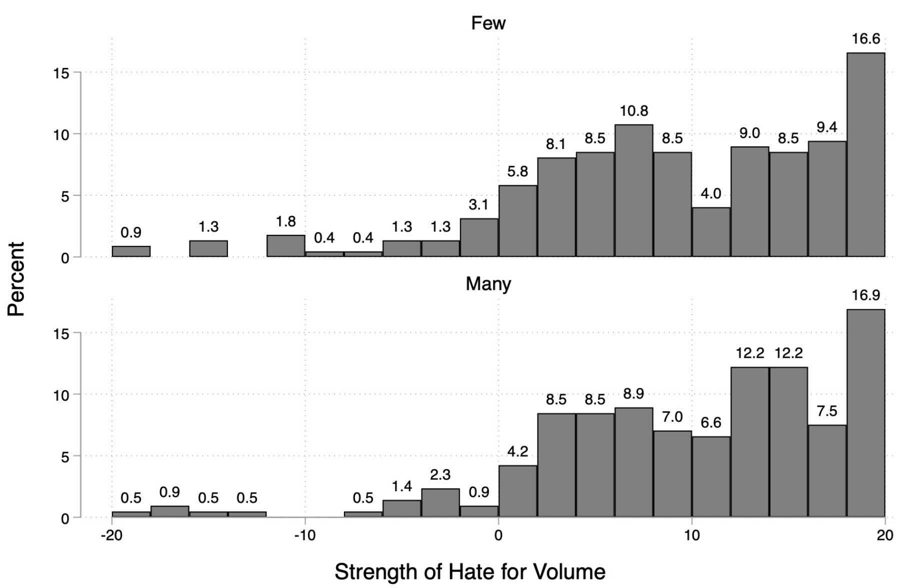

Figure B.4: Strength of Hate for Volume

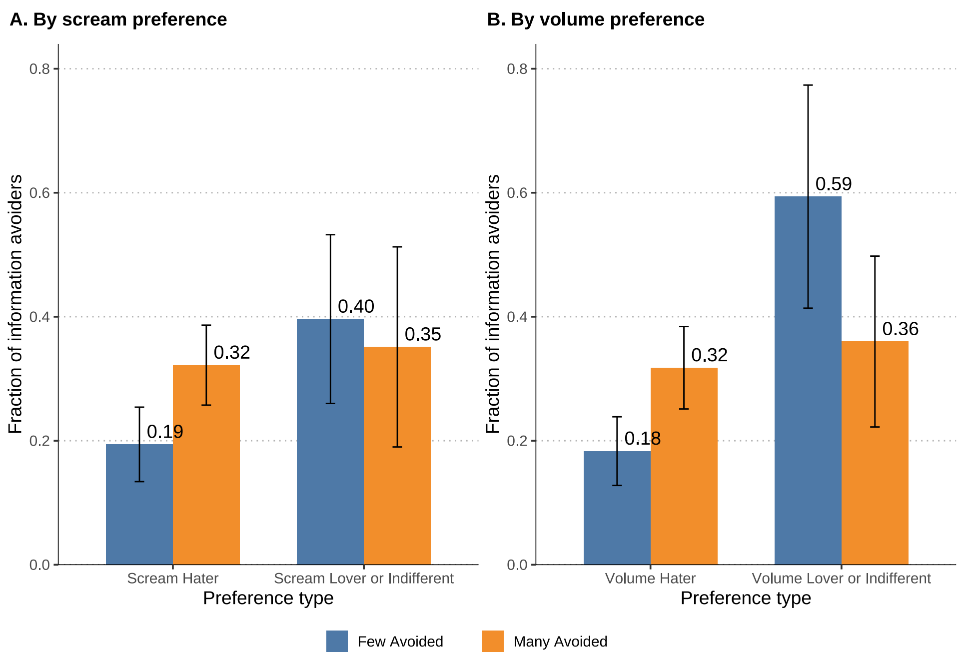

Figure B.5: Treatment effect on information avoidance by preference over screams and over volume

Notes:

	Dep. Var: Information Avoidance						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Many	0.128***	0.160***	0.156***	0.162***	0.179***	0.162***	0.169***
	(0.0448)	(0.0515)	(0.0510)	(0.0524)	(0.0519)	(0.0518)	(0.0518)
Age			0.0128*				0.0143*
			(0.00689)				(0.00729)
Female			−0.150***				−0.145***
			(0.0486)				(0.0524)
Patience					−0.0251		−0.0129
					(0.0264)		(0.0253)
Risk seeking					0.0165		−0.0134
					(0.0255)		(0.0265)
Pos. Reciprocity					−0.0329		−0.0307
					(0.0325)		(0.0318)
Neg. Reciprocity					0.0303		0.0331
					(0.0285)		(0.0283)
Altruism					−0.0378		−0.0482
					(0.0310)		(0.0311)
Trust					0.0443*		0.0273
					(0.0243)		(0.0248)
IPS score						−0.113*	−0.124**
						(0.0577)	(0.0586)
Constant	0.194***	0.291*	0.113	−0.0412	0.267	0.636**	0.132
	(0.0305)	(0.172)	(0.224)	(0.240)	(0.172)	(0.247)	(0.339)
Strata FEs		✓	✓	✓	✓	✓	✓
Field of study FEs				✓			✓
Observations	375	375	371	375	374	374	371

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.2: Main regression, sample of scream haters only

	Dep. Var: Information Avoidance						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Many	0.134***	0.153***	0.156***	0.159***	0.159***	0.153***	0.165***
	(0.0440)	(0.0494)	(0.0485)	(0.0501)	(0.0501)	(0.0494)	(0.0497)
Age			0.0131**				0.0169***
			(0.00631)				(0.00645)
Female			−0.157***				−0.137***
			(0.0467)				(0.0516)
Patience					−0.00879		−0.00776
					(0.0266)		(0.0250)
Risk seeking					0.0192		−0.00630
					(0.0234)		(0.0250)
Pos. Reciprocity					−0.0241		−0.0240
					(0.0304)		(0.0302)
Neg. Reciprocity					0.0202		0.0219
					(0.0272)		(0.0274)
Altruism					−0.0491		−0.0615**
					(0.0298)		(0.0299)
Trust					0.0185		−0.00266
					(0.0235)		(0.0240)
IPS score						−0.107*	−0.133**
						(0.0566)	(0.0568)
Constant	0.183***	0.0700	−0.129	−0.136	0.0689	0.396**	−0.0785
	(0.0281)	(0.0481)	(0.149)	(0.162)	(0.0476)	(0.185)	(0.284)
Strata FEs		✓	✓	✓	✓	✓	✓
Field of study FEs				✓			✓
Observations	383	383	380	383	383	383	380

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.3: Main regression, sample of volume haters only

C Supporting tables and figures

This section presents supporting tables and figures discussed but not shown in the body of the paper.

C.1 Descriptive statistics

	N	Mean	SD	Median	Min	Max
Age	460	22.509	3.713	22.000	18.000	47.000
Female	465	0.583	0.494	1.000	0.000	1.000
Field: Medicine	465	0.213	0.410	0.000	0.000	1.000
Field: STEM	465	0.200	0.400	0.000	0.000	1.000
Field: Humanities	465	0.228	0.420	0.000	0.000	1.000
Field: Economics	465	0.108	0.310	0.000	0.000	1.000
Field: Business	465	0.211	0.408	0.000	0.000	1.000
Field: Others	465	0.056	0.230	0.000	0.000	1.000
Work Experience	460	0.154	0.362	0.000	0.000	1.000
IPS score	464	3.116	0.414	3.111	1.000	4.000
Patience	464	-0.017	1.016	0.118	-3.749	1.174
Risk seeking	464	-0.014	0.998	0.233	-2.754	1.600
Pos. Reciprocity	464	-0.003	0.804	0.027	-4.006	1.331
Neg. Reciprocity	464	0.018	0.833	0.106	-1.918	1.908
Altruism	464	0.004	0.835	0.076	-2.203	2.231
Trust	464	0.003	0.997	-0.155	-1.662	2.082

Table C.1: Sample summary statistics

C.2 Information avoidance

Figure C.1 shows the estimated coefficients of a regression of information avoidance on individual characteristics in a linear probability model. The individual characteristics include age, gender, field of study, and economic and general information preferences as measured by the IPS (Ho et al., 2021).

Table C.2 reports the treatment balance checks.

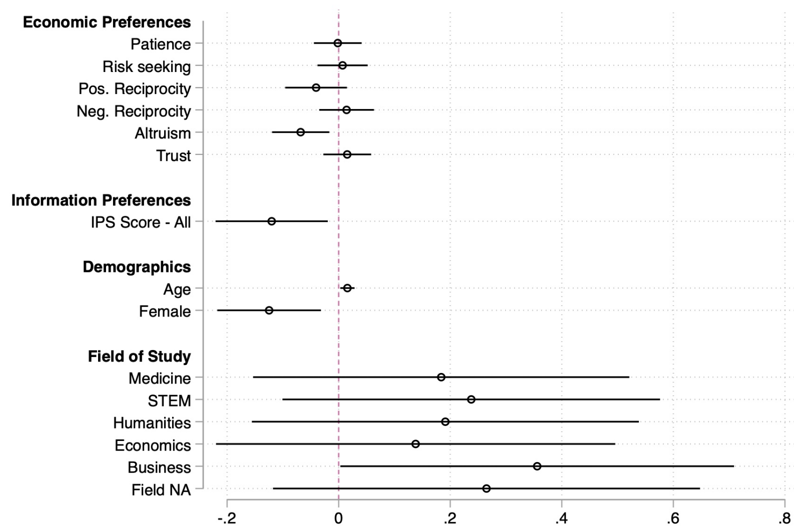

Figure C.1: Predictors of Information Avoidance

Notes: The figure shows the estimated coefficients of a regression of information avoidance on individual characteristics. The dependent variable is a binary indicator equal to one if the subject chose to avoid information and zero otherwise. The coefficients are estimated in a linear probability model. The error bars represent 95% confidence intervals. The regression includes all the characteristics shown in the figure. Economic preferences are measured by administering the relevant Global Preferences Survey questions (Falk et al., 2018). The information preferences are measured by the Information Preference Scale by Ho et al. (2021), with higher values indicating a higher propension to obtain information. The sample size is 465.

		(1) Few		(2) Many		(2)-(1) Pairwise t-test
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Age	220	22.532 (0.268)	240	22.488 (0.223)	460	-0.044
Female	223	0.570 (0.033)	242	0.595 (0.032)	465	0.026
Medicine	223	0.193 (0.026)	242	0.231 (0.027)	465	0.039
STEM	223	0.184 (0.026)	242	0.215 (0.026)	465	0.031
Humanistic	223	0.193 (0.026)	242	0.260 (0.028)	465	0.068*
Economics	223	0.139 (0.023)	242	0.079 (0.017)	465	-0.061**
Business	223	0.247 (0.029)	242	0.178 (0.025)	465	-0.069*
Other fields	223	0.063 (0.016)	242	0.050 (0.014)	465	-0.013
IPS Score - All	223	3.104 (0.028)	241	3.127 (0.026)	464	0.022
IPS Score - All (excl. occ.)	223	3.033 (0.031)	241	3.061 (0.027)	464	0.029
IPS Score - Health	223	3.042 (0.048)	241	3.082 (0.044)	464	0.040
IPS Score - Finance	223	2.885 (0.045)	241	2.900 (0.041)	464	0.016
IPS Score - Personal	223	3.111 (0.037)	241	3.139 (0.036)	464	0.027
IPS Score - General	223	3.045 (0.047)	241	3.079 (0.045)	464	0.034
IPS Score - Occupational	223	3.291 (0.032)	241	3.296 (0.032)	464	0.006

Table C.2: Balance of pre-treatment characteristics

D Robustness of treatment effect

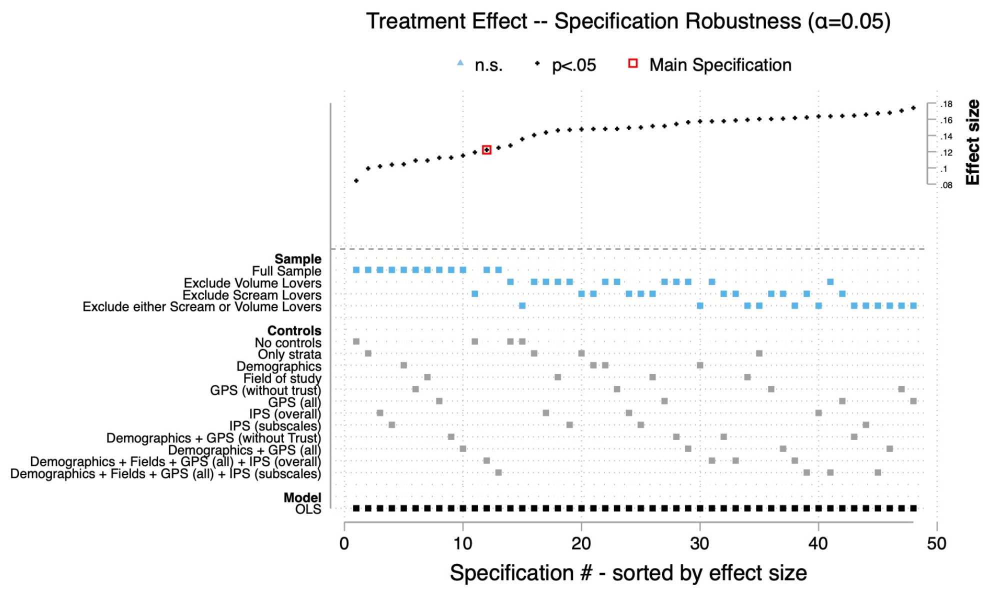

Figure D.1: Treatment effect robustness to alternative model specifications

Notes:

E Analyses of Information Avoidance Scale as Dependent Variable

[To be completed.]

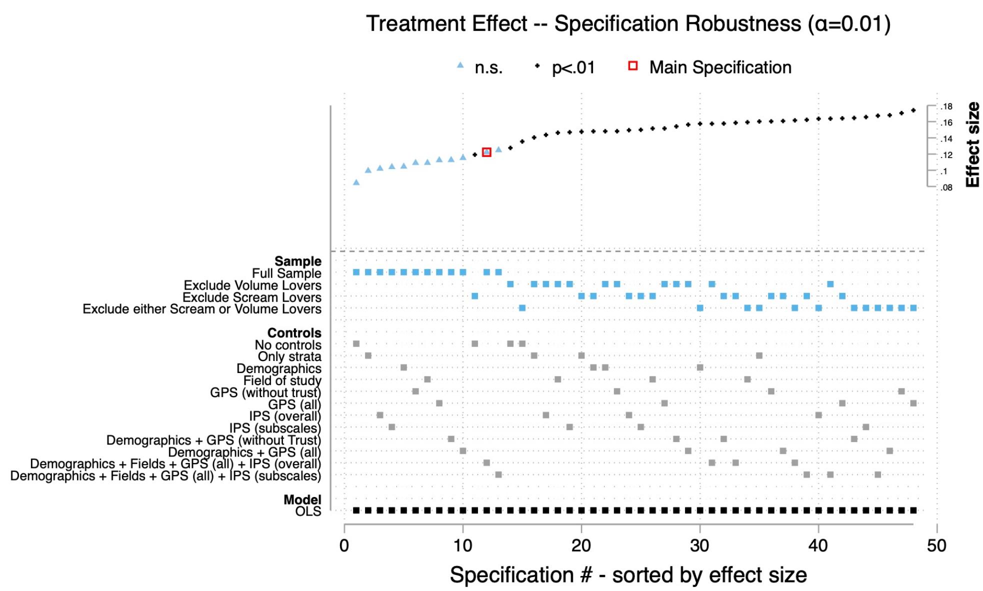

Figure D.2: Treatment effect robustness to alternative model specifications

Notes: